Grade 12 Mathematics of Data Management: MDM4U

Strands C, D, E

Theme: Decolonizing the Curriculum

Learning Task: 3-4 periods

Evaluation Task: 3-4 periods

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Expectations

Learning Task (Ontario Ministry of Education, 2007)

C1. demonstrate an understanding of the role of data in statistical studies and the variability inherent in data, and distinguish different types of data;

- C1.1 recognize and describe the role of data in statistical studies (e.g., the use of statistical techniques to extract or mine knowledge of relationships from data), describe examples of applications of statistical studies (e.g., in medical research, political decision making, market research), and recognize that conclusions drawn from statistical studies of the same relationship may differ (e.g., conclusions about the effect of increasing jail sentences on crime rates)

D1. Analyse, interpret, and draw conclusions from one-variable data using numerical and graphical summaries;

- D1.1 Determine the relevant numerical summaries for one-variable data (mean, median, mode, range, interquartile range, variance, and standard deviation).
- D1.5 interpret numerical and graphical summaries to describe the characteristics of a dataset and explain what the data shows

D2. Analyse, interpret, and draw conclusions from two-variable data using numerical, graphical, and algebraic summaries;

- D2.2 recognize and distinguish different types of relationships between two variables that have a mathematical correlation (e.g., the cause and-effect relationship between the age of a tree and its diameter; the common-cause relationship between ice cream sales and forest fires over the course of a year; the accidental relationship between the consumer price index and the number of known planets in the universe)
- D2.3 generate, using technology, the relevant graphical summaries of two-variable data (e.g., scatter plots, side-by-side boxplots) based on the type of data provided (e.g., categorical, ordinal, quantitative).
- D2.5 interpret statistical summaries (e.g., scatter plot, equation representing a relationship) to describe the characteristics of a two variable data set and to compare two related two-variable data sets (e.g., compare the relationship between Grade 12 English and mathematics marks with the relationship between Grade 12 science and mathematics marks); describe how statistical summaries (e.g., graphs, linear models) can be used to misrepresent two-variable data; and make inferences, and make and justify conclusions, from statistical summaries of two-variable data orally and in writing, using convincing arguments

E2. Communicate the findings of a culminating investigation and provide constructive critiques of the investigation of others

- E2.1 Compile a clear, well-organized, and detailed report of the investigation

Evaluation Task (Ontario Ministry of Education, 2007)

D1. Analyse, interpret, and draw conclusions from one-variable data using numerical and graphical summaries;

- D1.1 Determine the relevant numerical summaries for one-variable data (mean, median, mode, range, interquartile range, variance, and standard deviation).
- D1.5 interpret numerical and graphical summaries to describe the characteristics of a dataset and explain what the data shows

D2. Analyse, interpret, and draw conclusions from two-variable data using numerical, graphical, and algebraic summaries;

- D2.2 recognize and distinguish different types of relationships between two variables that have a mathematical correlation (e.g., the cause and-effect relationship between the age of a tree and its diameter; the common-cause relationship between ice cream sales and forest fires over the course of a year; the accidental relationship between the consumer price index and the number of known planets in the universe)
- D2.3 generate, using technology, the relevant graphical summaries of two-variable data (e.g., scatter plots, side-by-side boxplots) based on the type of data provided (e.g., categorical, ordinal, quantitative)
- D2.5 interpret statistical summaries (e.g., scatter plot, equation representing a relationship) to describe the characteristics of a two variable data set and to compare two related two-variable data sets (e.g., compare the relationship between Grade 12 English and mathematics marks with the relationship between Grade 12 science and mathematics marks); describe how statistical summaries (e.g., graphs, linear models) can be used to misrepresent two-variable data; and make inferences, and make and justify conclusions, from statistical summaries of two-variable data orally and in writing, using convincing arguments

E1. Design and carry out a culminating investigation that requires the integration and application of the knowledge and skills related to the expectations of this course.

- E1.5 draw conclusions from the analysis of the data (e.g., determine whether the analysis solves the problem), evaluate the strength of the evidence (e.g., by considering factors such as sample size or bias, or the number of times a game is played), specify any limitations of the conclusions, and suggest follow-up problems or investigations

E2. Communicate the findings of a culminating investigation and provide constructive critiques of the investigation of others

- E2.2 present a summary of the culminating investigation to an audience of their peers within a specified length of time, with technology (e.g. presentation software) or without technology

Learning Task

Land-Based Problem Options

Review and choose one of the following land-based problems to tackle for this assignment.

Option 1: Blanding's Turtle

> Blanding's turtle is an endangered species in Ontario that depends on wetland habitats and surrounding land to survive. The data focuses on population and habitat-related measures over time, allowing you to analyze trends and variability. This option includes fewer data points, which may make it easier to manage, but conclusions may be less certain.

Option 2: Emerald Ash Borer

> The emerald ash borer is an invasive insect that has spread through parts of Ontario, affecting ash tree populations in both urban and natural areas. The data tracks changes over time and across multiple regions, allowing you to examine patterns of spread, impact, and monitoring. This option provides more data to work with and may be more challenging, but strong analysis can lead to clearer and more well-supported conclusions.

After choosing your land-based problem, on a separate document or on paper, answer each of the following questions using the associated spreadsheet (Learning Task Datasets file). **Be sure to indicate which question you are answering.**

Google sheets learning task datasets file link:
https://docs.google.com/spreadsheets/d/1LT1xO9HcX-MhyW2TAmW4tiYwC4ctYSixNCzEs6sH5f0/edit?usp=sharing

Framing the Problem

1. Briefly describe your chosen issue and explain why it is relevant to Ontario's land and ecosystems.

2. After reviewing the data, write at least two clear, measurable questions you can ask to help guide your analysis?

Understanding the Data

3. Identify at least three important statistical calculations you will focus on and explain why each is useful for understanding the problem. (mean, median, range, variance, standard deviation)

4. Choose and justify at least two ways to organize or display the data (e.g., table, histogram, box-and-whisker plot, scatterplot). Explain why each representation is appropriate for your variables.

Statistical Analysis

5. Calculate the variables you identified in question 3. Show one worked example for each type of calculation you use.
6. Show your calculated data in a clear table. Label all columns and include units where appropriate.
7. Create at least two different graphs that support your argument (scatter plot, bar graph, histogram, boxplot, or pie chart). Explain why each graph is appropriate.

Interpreting the Evidence

8. Identify any trends, clusters, or outliers. Explain what they suggest about your topic.
9. What claims can you make about past, present, and future conditions based on your data? Clearly state any assumptions you make.

Discussing the Data

10. Discuss at least two strengths and two limitations of the dataset, and how they might affect your conclusions.

Reflection and Communication

11. Using what you have learned in this task, explain how the species in your chosen problem (invasive or at risk) could affect the survival of another species in the same ecosystem. Describe at least one way this relationship could cause a species to become at risk.
12. Reflect on what the data revealed that a simple description of the issue would not.
13. Write a brief summary of your findings so that anyone, regardless of math skill level, could understand your conclusions and evidence. Your summary should be between 200-400 words typed, or 3 pages handwritten.
14. Identify one area of statistics where your understanding could be improved.

Evaluation Task

In the learning task, you identified and justified why your chosen land-based problem is an issue. In this evaluation task, you will build on that work by justifying an appropriate restoration strategy. Below, you will find a table outlining two possible restoration strategies for each land-based problem (Blanding's Turtle and Emerald Ash Borer).

Your task is to review the two strategies for your selected problem and determine which one should be implemented. You are provided with data for each strategy; this data is representative of how similar strategies have been used at other affected sites in the past. You must use this evidence, along with the guiding questions below, to justify your recommended solution. Finally, you will present your argument using your choice of approved format (e.g., poster, video, presentation, conferencing etc.) so that a non-expert could understand.

Restoration Strategies	
Blanding's Turtle	**Emerald Ash Borer**
Habitat Expansion Habitat expansion involves increasing the total area of land and wetlands protected for Blanding's turtles. By providing more suitable habitat, this strategy aims to support larger and more stable populations.	**Removal and Replacement** This strategy focuses on removing ash trees that are already infected and replacing them with a variety of native tree species. The goal is to reduce the number of host trees available to the emerald ash borer while increasing forest diversity, which can make future infestations less severe.
Habitat Connectivity Habitat connectivity focuses on linking previously fragmented habitats through wildlife corridors. This allows turtles to move safely between areas for nesting, feeding, and overwintering.	**Containment Zones** Containment zones aim to slow or prevent the spread of the emerald ash borer by treating infected or high-risk areas with insecticides. Resources are concentrated where the risk of spread is greatest.

Google sheets evaluation task dataset file link:
https://docs.google.com/spreadsheets/d/1HPAlpEUnvvWw_pICasC2z0d_QXUui2SAjvHWTRVawGs/edit?usp=sharing

On a separate document or on paper, answer each of the following questions using the associated spreadsheet (Evaluation Task Datasets file). **Be sure to indicate which question you are answering.**

Framing the Problem

1. How could the effectiveness of each strategy be measured?
2. How could you compare the effectiveness of each strategy?

Understanding the Data

3. Identify the variables you will use to compare the two strategies. Explain why these variables are relevant.

Statistical Analysis

4. Calculate appropriate statistical measures (e.g., mean, median, range, IQR, standard deviation) for each strategy. Show **one** fully worked example for each type of calculation used.
5. Create at least two graphs that support a comparison of the strategies (scatter plot, bar graph, histogram, boxplot, or pie chart). Explain why each graph is appropriate.

Interpreting the Evidence

6. Using statistical evidence, compare the short/long term outcomes of each strategy.
7. Identify any important trade-offs or real-world constraints (e.g., cost, variability, time, practicality) shown in the data.

Discussing the Data

8. Discuss at least two strengths and two limitations of the data used in your comparison.
9. Based on the data, determine which strategy appears to be more effective or sustainable. Support your decision with statistical evidence.
10. Suggest at least one follow-up investigation or additional data that could improve future decision-making.

Reflection and Communication

11. What main points will be essential to convey your message? Use these points to build your presentation.

12. After completing your presentation of findings, reflect on this task and describe one thing that went well, one thing that challenged you, and one thing you would have done differently if given the option to try again.

Evaluation Task Rubric

Criteria	Level 1 (50–59%)	Level 2 (60–69%)	Level 3 (70–79%)	Level 4 (80–100%)
Understanding of Restoration Strategies (Knowledge and Understanding, Application) **10 marks**	Demonstrates a limited understanding of the two strategies. Purpose and goals are unclear or inaccurate.	Demonstrates a partial understanding of both strategies, with some minor misunderstandings.	Demonstrates a clear and accurate understanding of both strategies and what each aims to improve.	Demonstrates a thorough understanding of both strategies, including how and why they differ in approach and goals.
Selection and Use of Data (Thinking) **5 marks**	Uses very little relevant data or relies on inappropriate relationships. Data selection is not justified.	Uses some relevant data, but selection may be incomplete or weakly justified.	Selects appropriate variables and explains why they are relevant to comparing strategies.	Thoughtfully selects multiple relevant variables and clearly justifies how each contributes to evaluating strategy effectiveness.
Statistical Analysis (Knowledge and Understanding) **15 marks**	Statistical measures are missing, incorrect, or inappropriate. Graphs are unclear or not meaningful.	Some appropriate statistical measures and graphs are used, but with errors or limited explanation.	Appropriate statistical measures and graphs are correctly calculated and clearly presented.	Statistical measures and graphs are accurate, well-chosen, and clearly linked to the research question, showing strong analytical judgment.
Interpretation of Evidence (Application) **15 marks**	Interpretations are vague, inaccurate, or not supported by data.	Interpretations reference data but may be superficial or partially unsupported.	Interpretations clearly connect trends, variability, and statistics to the comparison of strategies.	Provides insightful interpretation of trends, variability, and projections, demonstrating deep understanding of what the data shows and implies.

Justification of Chosen Strategy (Application) **15 marks**	A strategy is chosen with little or no supporting evidence.	A strategy is chosen with some evidence, but the argument lacks clarity or depth.	A clear strategy is justified using relevant statistical evidence and comparisons.	A compelling, well-structured argument clearly supports one strategy, addressing both short- and long-term outcomes and trade-offs.
Evaluation of Data Strengths and Limitations (Application) **5 marks**	Little or no discussion of data strengths or limitations.	Identifies some limitations or strengths, but discussion is brief or unclear.	Clearly discusses strengths, limitations, and variability, and explains how these affect confidence in conclusions.	Thoughtfully evaluates data quality, bias, variability, and uncertainty, showing strong critical thinking.
Follow-Up Investigation (Application) **5 marks**	Follow-up suggestions are missing or unrealistic.	Follow-up investigation is suggested but lacks detail or relevance.	Suggests a reasonable and relevant follow-up investigation or additional data.	Proposes a well-reasoned follow-up investigation that clearly addresses limitations or unanswered questions.
Communication and Presentation (Communication) **20 marks**	Communication is unclear, or difficult to follow. Presentation is disorganized or poorly suited to the chosen format.	Some effort to communicate clearly.. Organization is inconsistent or format is only partially effective.	Information is clearly organized, the format is appropriate, and information is explained in a way that all math skill levels can understand.	Communication is exceptionally clear, engaging, and accessible. Presentation is polished, well-structured, and makes excellent use of the chosen format.

Learning Task Solutions

Solutions will vary between students depending on the topic they choose, and the variables they analyse. To create the best possible guide for educators, an example solution has been provided for each topic, along with feedback.

Student Responses are in black, and teacher feedback is in red.

Blanding's Turtle Solution

Framing the Problem

1. Briefly describe your chosen issue and explain why it is relevant to Ontario's land and ecosystems.

Selected Option 1

The Blanding's turtle is native to Ontario, and its habitat extends from all of Southern Ontario to North of Timmins (Government of Ontario, 2020). They are both omnivores and scavengers, supporting their habitat's food web, recycling nutrients, and controlling insect and algae populations (Georgian Bay Mnidoo Gamii Biosphere, 2025). The Blanding's turtle also plays a significant role in Anishinaabek culture, as it has ties to Anishinaabe teachings and Creation stories, functions as a natural calendar, and plays an important role in the Anishinaabe clan systems (Georgian Bay Mnidoo Gamii Biosphere, 2025).

2. Write at least two clear, measurable statistical questions that can be answered using the provided data.

Question 1: Is there a relationship between the wetland area (km^2) and the estimated population of Blanding's turtle population in Southwest Ontario from 2010 to 2026?

I like how you ask your questions without a bias towards a particular strategy.

Question 2: Is there a relationship between average annual temperature (°C) and nest success rate (%) in Southwest Ontario from 2010 to 2026?

Understanding the Data

3. Identify at least three important variables you will focus on and explain why each is useful for understanding the problem. (mean, median, range, variance, standard deviation)

Variable 1: Estimated population

The estimated population is an important variable because it shows whether the turtle's population is increasing or decreasing. For this variable I could calculate the mean to see how

much each year the population deviated away from the average. I could also calculate the variance or the standard deviation to see how much the population changes year to year, and if it is stable or inconsistent.

Variable 2: Wetland Area (km^2)

The wetland area (km^2) is an important variable because this is the habitat where turtles live. We can look at trends in the amount of available habitat to see if less available habitat means there are fewer turtles (and the other way around).

I like how you made the connection between a potentially shrinking wetlands as being a driving factor for the Blanding's turtle endangerment. I would have liked to see you correlate this to habitat fragmentation, as the two often go hand in hand.

Variable 3: Average Annual temperature (°C)

The average annual temperature is an important variable because it is an environmental condition that could be linked to habitats of turtles. Temperature from my experience is something that changes with seasons and warmer springs or colder winters could affect reproduction. I could use the range and standard deviation to find out how much temperature changes year to year and then compare that to the nesting success rates to see if warmer / colder years affect nesting success rate.

Variable 4: Nesting Success Rate (%)

The Nesting Success Rate is important because it shows how successful turtles are at reproducing each year. If nest success is low, fewer baby turtles will survive, which can affect the population later on. For this variable I could use the mean and the median to find an estimate of what it should be and then using range and standard deviation to see if nest success is steady or if there are some years that are way higher or lower than normal.

4. Choose and justify at least two ways to organize or display the data (e.g., table, histogram, box-and-whisker plot, scatterplot). Explain why each representation is appropriate for your variables.

I could use a scatter plot for both of my questions. For my first question, a scatterplot can be made by putting the wetland area on the x-axis and the estimated population on the y-axis. From this I can see if there is a trend from the data by potentially drawing a line of best fit. For my other question I could use a scatter plot in the same way to see if there is a positive or negative correlation between the nesting success rate and temperature changes.

Your explanation is detailed and connects to your questions.

A table could be another way to put the data together. Although the data is already given to us in a table, by creating a table that focuses on the selected variables it allows for easier comparison of the data to analyze trends.

Using the graph to represent data is sufficient.

A histogram would be useful to show if there are repeat values / where values tend to lie closer to. If you were to divide the data into quartiles then group based on which values fall into which quartiles then you could create a histogram that reflects which values are most common.

Implementing a histogram can be difficult in this context, but as you describe it, this would work.

Statistical Analysis

5. Calculate the variables you identified in question 3. Show one worked example for each type of calculation you use.

Variable 1 Calculation for the estimated population

Mean = sum of all population values divided by the number of data entries

Mean = 7531 / 17

Mean = 443

Median calculation for the population (finding the middle value)

For doing this in google sheets what I did was organize the data from largest to smallest then I counted to the middle.

380, 388, 395, 402, 410, 418, 425, 432, **440**, 448, 455, 460, 470, 485, 498, 505, 520

Range Calculation

For finding the range what I did was take the largest value in the data set from the population then subtracted it by the smallest value to get the difference. The difference between the two values is going to be the range of the data set (the top and bottom).

Range = max - min

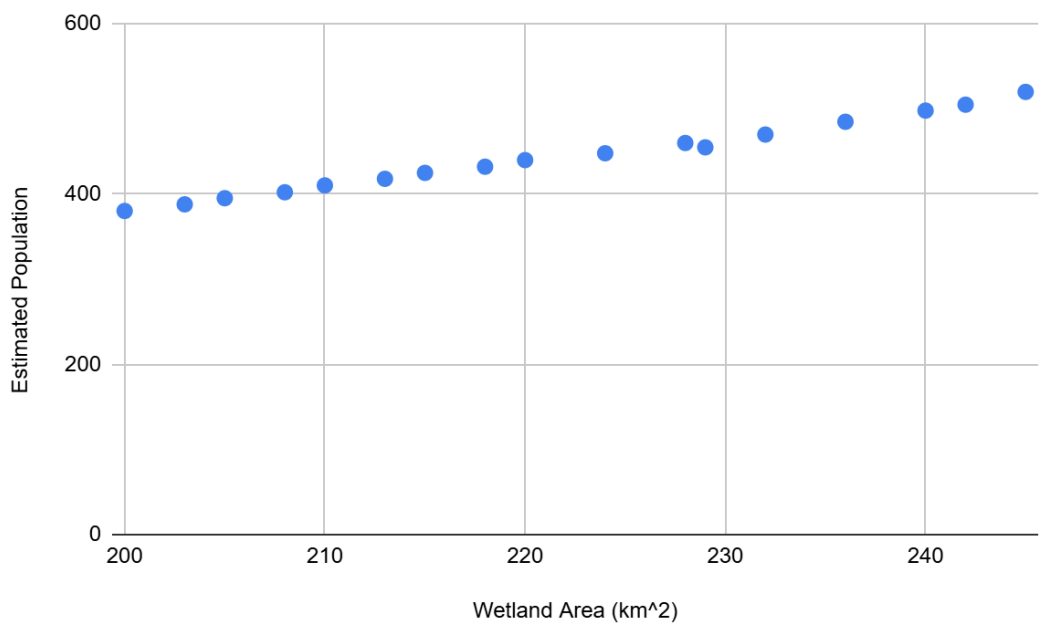
Range = 520 - 380

Range = 140

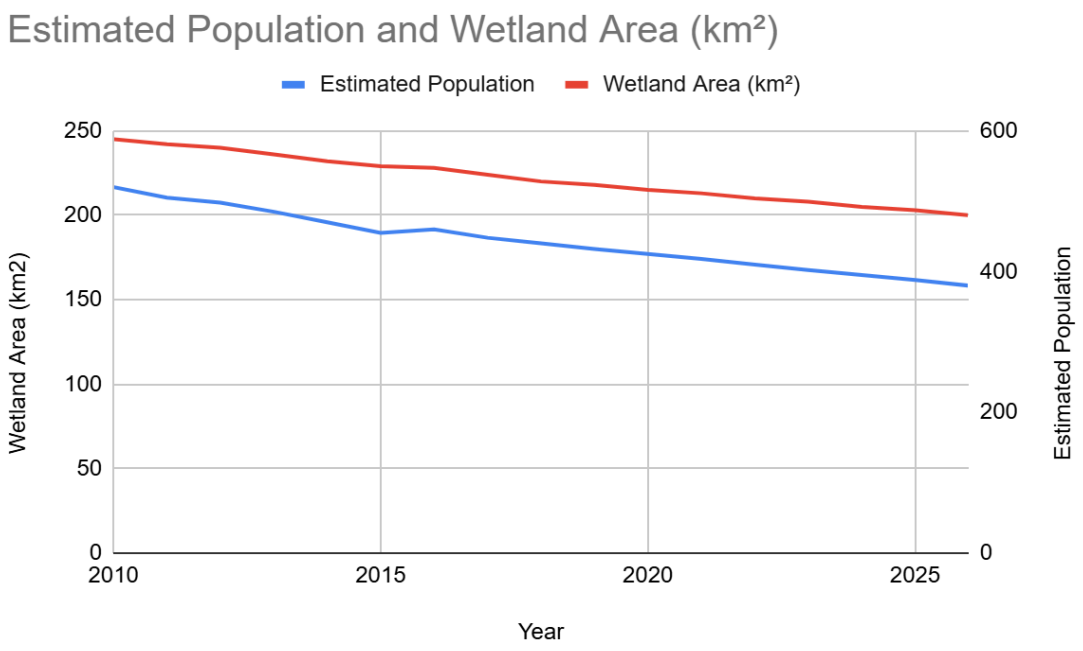
6. Show your calculated data in a clear table. Label all columns and include units where appropriate.

Variable	n	Mean	Median	range	Variance	Std dev
Estimated Population	17	443.00	440.00	140.00 turtles	1806.00 turtles²	42.50 turtles
Wetland Area	17	221.65	220.00	45.00 km²	203.74 (km²)²	14.27 km²
Nest Success Rate	17	50.94	50.00	20.00 %	40.31 %²	6.35 %
Avg Annual Temperature	17	9.14	9.20	2.20 °C	0.46 (°C)²	0.67 °C

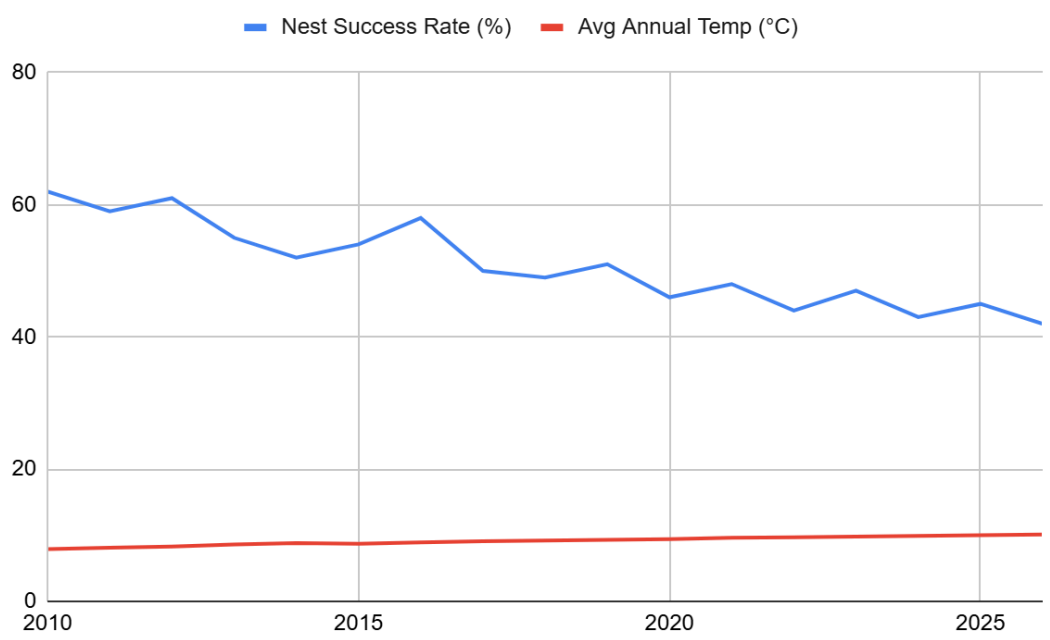
7. Create at least two different graphs that support your argument. Explain why each graph is appropriate.



Don’t forget your graph titles.



For the above graph here, I think this graph is appropriate for showing the relationship between wetland area and estimated turtle population because the x axis is the independent variable and the dependent variable is the y axis. Since we want to see how the size of the wetland area affects the population we can see that for bigger wetland area values we get a bigger estimated population value.



This graph is an appropriate choice because we are able to see how two variables over time change. For the first variable we have nest success rate which we can see is showing a decreasing trend from 2010 to 2026. Then for the second trend line (the average annual temperature) we can see that it is slightly increasing. This allows us to see that as temperature increases over the year the nesting success rate decreases.

How do you know the nest success rate and annual temperature are correlated? How does the graph indicate a correlation other than just, as one increases the other decreases?

Interpreting the Evidence

8. Identify any trends, clusters, or outliers. Explain what they suggest about your topic.

Looking at the graphs one trend I can see in the wetland area and population graph is that as the wetland area increased so did the estimated population value. This to me suggests that with more habitat available there is a larger capacity to hold more turtles.

Another trend that I noticed was that as the temperature increased over the years the nesting success rate decreased. This to me suggests that warmer average annual temperature makes it increasingly difficult to create favorable nesting conditions for turtles. This could have an impact on the wetland habitat directly making it difficult for the turtles to live there.

When it comes to the clusters for the data, there are years where there is not much change. For example looking at the graph, although it is difficult to notice it here, there is not much of a change in temperature from 2010-2015.

9. What claims can you make about past, present, and future conditions based on your data? Clearly state any assumptions you make.

When it comes to the past data (referring to the years 2010 to 2014), the data shows that more wetland area, a higher estimated turtle population, cooler average temperatures, and higher nest success rates compared to the later years. Fast forward to the present years (referred to the years 2024 to 2026) the data shows less wetland area, a lower estimated turtle population with warmer average temperatures, and lower nesting success rates. When comparing the past to the present, if the trend continues, the data suggests that the turtle population may keep decreasing, wetland area may continue to shrink, and nest success may stay lower or keep dropping as temperatures remain higher. Assuming this trend continues in a similar pattern would imply that we would be assuming a similar trend in conditions in the future. However, weather patterns could get worse non linearly, and we could also be losing wetland area at greater intervals each year.

Could you use linear regression to estimate the future outcomes for the Blanding's turtle?

Discussing the Data

10. Discuss at least two strengths and two limitations of the dataset, and how they might affect your conclusions.

One strength of the dataset is that it covers a 17 year period which is a long enough period of time to notice a trend in the data. The 17 year sample size of the data shows a more recent picture of the trends which can make it easier to see the data in the short term future. If we were to include data from 100 years back that could make it difficult to reflect the short term effects since we would most likely be including points that don't really show a change. Additional data points could introduce more outliers to the dataset as well which can skew data as well. Another strength of the dataset is that the variables in the table represent the issue and how different parts of it can be analyzed to see how it affects it overall. By having different factors that could affect the habitat of the turtle over time we are able to get a better understanding of what could be impacting the turtles specifically. Knowing what specific variables are having an impact on the turtles could clarify what needs to be done for the turtles to help with the sustainability of their population and numbers. A limitation of the dataset is that the information is limited to just southwest Ontario. Although we are located in Ontario, it could be difficult to make conclusions for Blanding's turtles present in other parts of the country or even the world as there may be different factors affecting the habitat. Additionally, the variables and the trends in the variables for southwestern Ontario may be different than those of the variables in other areas. Another limitation in the data is the data itself for there are no references to how it was sourced. For example, if we are not able to see the method of which turtle populations were estimated through then how are we certain that those estimates are accurate reflections. Uncertainty in the methods used for estimating makes it difficult to draw conclusions from the data.

I like how you talked about more than just the data itself, pointing out the importance of reliable data.

To improve the reliability of my results, I would want more detailed data collected from multiple wetlands or locations each year, instead of one regional value. This would make the results more representative and reduce the impact of outliers. It would also help to collect data more frequently, such as seasonal temperature and nesting information, because nesting success is affected by specific conditions during the nesting period, not just the yearly average. Finally, adding other variables that could affect turtles would strengthen the analysis, such as road density or traffic volume near wetlands, predator levels, water quality, and information about conservation actions or habitat restoration happening in certain years.

How would you use the added variable, such as water quality, to support your claims?

Reflection and Communication

11. Using what you have learned in this task, explain how the species in your chosen problem (invasive or at risk) could affect the survival of another species in the same ecosystem. Describe at least one way this relationship could cause a species to become at risk.

Blanding's turtles are an important indicator for the health of wetland ecosystems. Within these ecosystems the turtles are a part of the foodweb and live with other species in the same habitat. A decline or increase in the turtle populations could impact other species through food web interactions. For example if there are less turtles in the ecosystem then the prey of the turtles may see an increase in their number, since there is an increase in their number they begin to surpass their threshold of resources in that habitat. This increase in other species causes an imbalance where some species such as the prey of turtles increase and begin to compete with other species in the ecosystem for resources. This kind of relationship can cause another species to become at risk if it leads to long-term declines in survival or reproduction. For instance, if changes in the wetland reduce frog breeding sites or increase disease and poor water conditions, fewer frog eggs and tadpoles survive each year. If that happens consistently, the frog population could keep shrinking until it becomes threatened or at risk. Another way this can happen is if the same human impacts harming turtles, like wetland loss and road mortality, also reduce habitat for other wetland species, causing multiple populations to decline at the same time.

Your argument would be much stronger if you included references to reputable sources.

12. Reflect on what the data revealed that a simple description of the issue would not.

A simple description of the issue might say that Blanding's turtles are declining because their habitat is being impacted, but the data showed how consistent and connected the changes are over time. From 2010 to 2026 in Southwest Ontario, the estimated population dropped from 520 to 380 turtles, while wetland area also decreased from 245 km² to 200 km². Seeing these values across 17 years matters because it suggests the decline is not caused by one unusual year, but by a longer pattern. The data also showed that nest success did not just "get worse" in a vague way. It generally decreased from about 62% to about 42%, and the year-to-year variation makes it clear that some years are better than others, so it is important to look at overall trends rather than focusing on one year. Temperature increased over the same period, from about 7.9°C to 10.1°C, which gives evidence that environmental conditions are changing at the same time that nest success is lower.

Yes, an argument is far stronger when supported by quantitative data.

13. Write a brief summary of your findings so that a non-expert can understand your conclusions and evidence? Your summary should be between 200-400 words typed, or 3 pages handwritten.

From 2010 to 2026 in Southwest Ontario, the data shows that the Blanding's turtles population has been declining over time, and this decline is happening alongside changes to their habitat and environmental conditions. The estimated turtle population dropped from 520 turtles in 2010 to 380 turtles in 2026. During the same time period, the amount of wetland habitat also decreased from 245 km² to 200 km². Because Blanding's turtles rely on wetlands to find food, shelter, and nesting areas, this loss of habitat likely makes it harder for the population to stay healthy. The

data also shows changes that could affect reproduction. The nest success rate, which is the percentage of nests that successfully produce young, decreased from about 62% in 2010 to about 42% in 2026. Although nest success does go up and down from year to year, the overall pattern shows lower success in most recent years. At the same time, the average annual temperature increased from about 7.9°C to about 10.1°C. This matters because temperature can affect wetland conditions and nesting environment, which may be linked to nest success.

With the support from graphs (please refer to previous parts), I was able to see these trends clearly. The trends showed that population and wetland area both decreased over time, while temperature increased. Scatterplots can also be used to check whether wetland area and population move together, and whether higher temperatures tend to be linked with lower nest success. Overall, the evidence suggests that habitat loss and changing environmental conditions are happening at the same time as the turtle population decline, which could be contributing to the species becoming more at risk.

I like how you don't explicitly say that these variables are causing the decline in population, but instead indicate that they are likely contributing factors. I think your argument would be stronger if you mentioned the rate at which the turtle population is declining and when the turtle population can be expected to go extinct in the area. Including this would exemplify the urgency for action.

14. Identify one area of statistics where your understanding could be improved.

One area of statistics where I could improve is using correlation and regression more confidently. I understand that scatterplots can show relationships, but I want to get better at calculating and interpreting correlation values, using a line of best fit appropriately, and explaining what the strength of a relationship means without making causal claims.

I agree, all of these points would make your arguments stronger. Try looking at the slope of some of your lines of best fit, they may help you correlate your data.

Emerald Ash Borer Solution

Student Responses are in black, and teacher feedback is in red.

Framing the Problem

1. Briefly describe your chosen issue and explain why it is relevant to Ontario's land and ecosystems.

The emerald ash borer is a beetle, invasive to North America, which feeds on ash trees during their larvae stage. Although the emerald ash borer can fly up to 10 kilometers, it primarily travels through the movement of firewood and other infested ash wood products (Government of Canada, 2025).

Good succinct description to gain some insight into the problem. I like the included citation as well to show where you got information from.

2. After reviewing the data, write at least two clear, measurable questions you can ask to help guide your analysis?

    1) Is there a relationship between a region's total ash trees and infested trees?
    2) Does the average (mean) of the annual increase in infestation distance and sites correlate to the average annual canopy loss and ash tree loss?

Questions are good and can be tested with the use of graphs and statistics. One thing to consider is are you planning on looking at comparing the data for all years provided or just specific timeframes.

Understanding the Data

3. Identify at least three important statistical calculations you will focus on and explain why each is useful for understanding the problem. (mean, median, range, variance, standard deviation)

The mean will be important because it will give an indication of how the emerald ash borer affects ash trees on average. This will allow for a more accurate representation of the overall effects of the emerald ash borer, rather than looking at its effect on one particular region.

Standard deviation of the control region (Hudson BayLowlands) will give an indication as to whether or not the change in population for the other regions will be significant or not.

Range will be useful for comparing the extremes of the control region (Hudson Bay Lowlands), the late infestation region (Thunder Bay), and the early infestation region (GTA). A larger difference between ranges will indicate significance in the data.

Overall good explanations that are clear but be careful with the standard deviation. Standard deviation should describe how much values vary within each region (and you can compare regions to the control), and range shows spread but does not prove “significance” on its own, it just highlights extremes and variability.

4. Choose and justify at least two ways to organize or display the data (e.g., table, histogram, box-and-whisker plot, scatterplot). Explain why each representation is appropriate for your variables.

Average changes can be represented with a scatter plot, with linear trend lines which can have their slopes compared to identify a correlation in changes between the data. The closer the slopes, the larger the correlation. The average changes can also be represented with a line graph for easier, and faster review of data.

A bar graph can be used to display the range of data for each region's total ash trees and infested trees. A side by side comparison of this data can show to exemplify the impact the emerald ash borer has on an ash tree population.

Your choice of scatterplots/line graphs and comparison bar graphs fits the variables well and clearly supports both trend analysis over time and differences between regions. To build on the scatterplot idea, how do you think you could apply linear regression to model the relationship between the two variables and interpret the strength of that relationship?

Statistical Analysis

5. Calculate the variables you identified in question 3. Show one worked example for each type of calculation you use.

Calculations for Control Group

mean (total population)

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$$

$$= \frac{1}{16} \sum_{i=1}^{16} x_i$$

$$= 128111$$

Standard Deviation (total population)

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

$$= \sqrt{\frac{\sum (x_i - 128111)^2}{16}}$$

$$= 1681$$

Percentage (population lost)

$$\% \text{ lost} = 100\% - 100\% (\text{min} \div \text{max})$$

$$= 100\% - 100\% (128000 - 131000)$$

$$= 2\%$$

Range (total population)

Range = Max - Min

= 131000 - 128000

= 3000

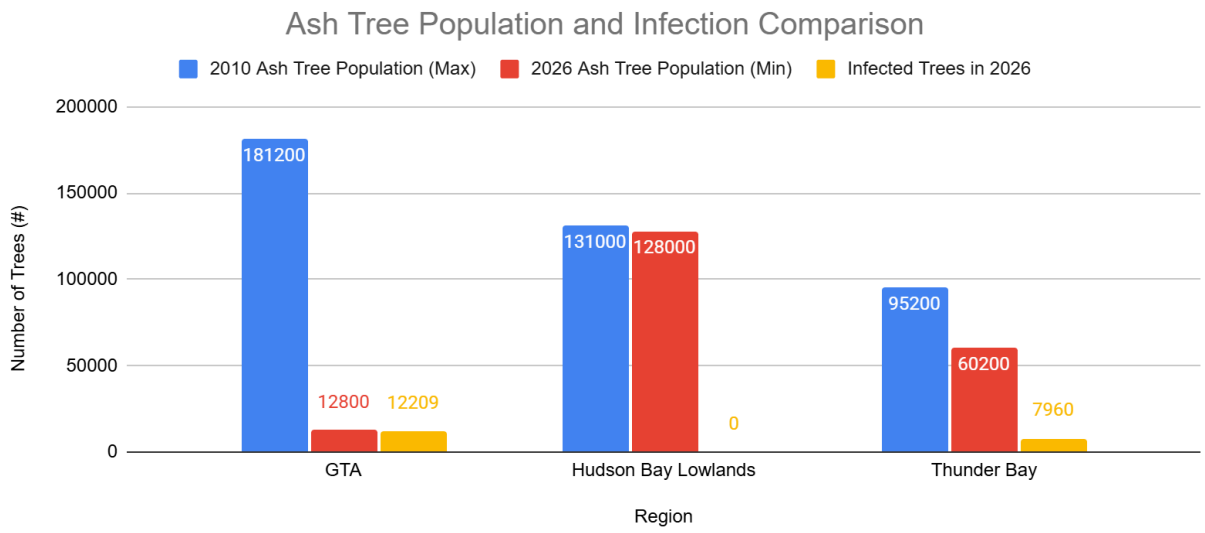
Calculations are good but I would recommend in the future to show calculations with other variables as well instead of showing for one variable.

6. Show your calculated data in a clear table. Label all columns and include units where appropriate.

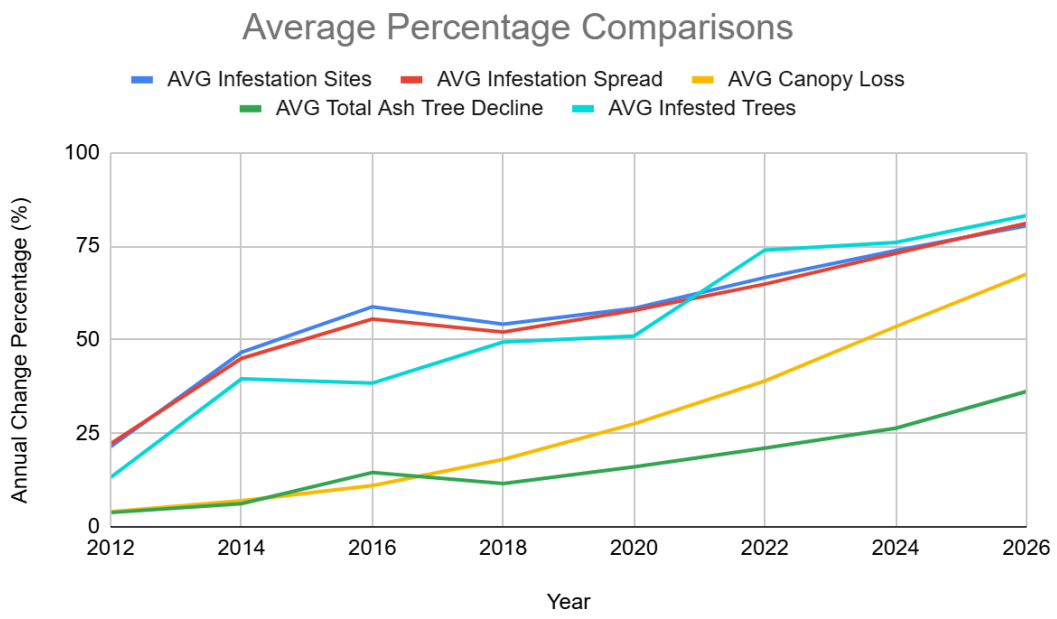
Region	Ash Trees Lost (Range)	Range of Infested Trees	2010 Ash Tree Population (Max)	2026 Ash Tree Population (Min)	Infected Trees in 2026
GTA	168400	11309	181200	12800	12209
Hudson Bay Lowlands	3000	0	131000	128000	0
Thunder Bay	35000	7960	95200	60200	7960

Year	Infestation Change (%)	Infestation Spread (%)	Mean Infestation Sites	Mean Infestation Spread	Mean Canopy Loss	Mean Total Ash Tree Decline	Mean Infested Trees
2012	21	22	21	22	4	4	39
2014	47	45	47	45	7	6	62
2016	59	56	59	56	11	14	72
2018	75	68	54	52	18	12	58
2020	74	64	58	58	28	16	72
2022	78	69	67	65	39	21	79
2024	84	77	74	73	54	26	83
2026	90	86	80	81	68	36	85

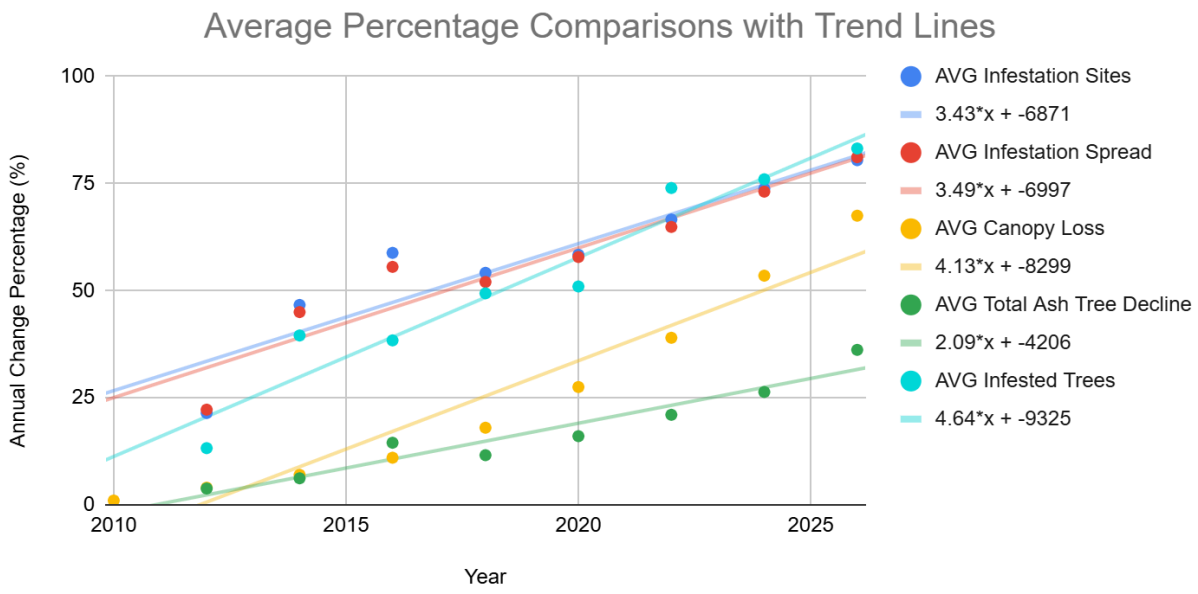
7. Create at least two different graphs that support your argument. Explain why each graph is appropriate.



This graph shows the initial and final ash tree population, as well as the number of infected trees in 2026. This graph is important because it shows how populations change in response to the emerald ash borer.



The graph shows the correlation between the average number of infestation sites, infestation spread, and total ash tree decline. It also shows the volatility in the average number of infested trees, and how the canopy loss increases more than the loss in ash trees, suggesting other species are dying off as well.



This graph uses the same data as the previous one, but instead takes the form of a scatter plot with linear trendlines. This graph shows the close correlation between increasing infestation sites, increasing infestation spread, and the increasing loss in the total ash tree population.

These are good choices of graphs, however for the third graph it may be easier to focus on fewer variables to help make information more clear.

Interpreting the Evidence

8. Identify any trends, clusters, or outliers. Explain what they suggest about your topic.

The average increase in infested trees followed a similar increase as the number of infestation sites and infestation spread, but with greater validity. This may be due to the fact that each tree, rather than a region, would have to be accurately measured, and as a result allowed for a larger margin of error and volatility.

Good job identifying the similarities in the trend between infested trees, sites and spread. Outside of this trend here did you notice any clusters within the data or any outliers present? If there are outliers or clusters in the data what may have caused them? On the other hand if there are no clusters or outliers how do you think this affects the data?

9. What claims can you make about past, present, and future conditions based on your data? Clearly state any assumptions you make.

The data suggests that, in the past, the GTA region had the greatest number of ash trees (181200), followed by the Hudson Bay Lowlands (131000), then Thunder Bay (95200). After the GTA region has been infected over 16 years, it now has the lowest population (12800), followed by Thunder Bay (60200) which has been infected at this point for 10 years, and the Hudson Bay

Lowlands, which has not been infected, has the largest population of (128000). If these trends continue, the infected regions (GTA and Thunder Bay) will lose the entirety of the ash tree population. As long as the Hudson Bay Lowlands doesn't get infected, it is to be expected to remain around 128111 trees, with a standard deviation of 1681 trees.

I like how you stated your claim in regards to the past, present and future conditions based on the data but is there any calculation (or representation) you used that could have helped with supporting these claims? I would like to see a reference to a calculation performed or a graph (if applicable here) to help support your claims.

Discussing the Data

10. Discuss at least two strengths and two limitations of the dataset, and how they might affect your conclusions.

The dataset includes a wide range of variables to work with, allowing for a large number of comparisons.

The dataset includes 3 different diverse regions, an already infected region, a region infected later on, and a control region which is never infected. This gives a dynamic view of how the ash trees are affected at different stages of infection.

The dataset only includes 3 regions, where only one of these is not infected. Having more infected and non-infected regions would strengthen any arguments.

The dataset doesn't include any factors that could affect the trees other than the emerald ash borer. There could be outside factors affecting the trees, for example extreme weather conditions.

Reflection and Communication

11. Using what you have learned in this task, explain how the species in your chosen problem (invasive or at risk) could affect the survival of another species in the same ecosystem. Describe at least one way this relationship could cause a species to become at risk.

The increase of an invasive species can cause the population of a native species to decrease. The introduction of an invasive species doesn't balance out either, and can cause a native species to die out entirely, which appears to be what is happening to the ash trees due to the emerald ash borer. With this risk present, species such as the emerald ash are at risk.

12. Reflect on what the data revealed that a simple description of the issue would not.

The data showed the correlation between the increase in the invasive emerald ash borer and the decrease in the ash trees. The data showed that the loss of ash trees in infected areas is far worse than a simple description conveyed. The canopy loss data also showed how more than just the

ash trees are experiencing a loss, as the canopy density is decreasing faster than the ash trees are dying off.

13. Write a brief summary of your findings so that anyone, regardless of math skill level, could understand your conclusions and evidence. Your summary should be between 200-400 words typed, or 3 pages handwritten.

Looking at the Average Percentage Comparisons chart, you can see that the average (mean) spread of the infestation, the increase in infestation sites, and the total ash tree population decline follow a very similar trend. The Average Percentage Comparisons with Trend Lines chart confirms this, showing that the variables are all changing at a very similar rate (slope difference of less than 1.34). This chart also shows that there's a greater loss in the forest canopy than there is with just the ash trees. This suggests that the emerald ash borer may be affecting more than just the ash trees.

The Ash Tree Population and Infection Comparison chart shows the change in the ash tree population and the change in the number of infested trees for different regions. The region that was already infected (GTA) showed a loss of 168,400 ash trees over a 16 year period (93% of the population). The region that was not infected (Hudson Bay Lowlands) experienced a minor loss of 3000 trees (2% of the population). The region infected half way through experienced a loss of 35000 trees (59% of the population). The infected areas both experienced a loss far greater than would be expected for a non-infected population, suggesting that the emerald ash borer is causing a concerning amount of tree loss.

The Hudson Bay Lowlands population comparison suggests that, without the emerald ash borer, the tree population should remain relatively stable around 128111 trees, losing or gaining around 1681 trees a year. From the GTA data we can conclude that when a large percentage of the ash tree population is infected, the population tends to die off. The Thunder Bay population confirms this, as its population is decreasing as the number of infected trees is positive. If this were allowed to go on without intervention, the entire ash tree population in infected regions will eventually die off.

I like how you included specific values and other measures in your explanation. Additionally it was nice to see the predicted outcome and what you took away from the data in the conclusion. In the future I may include a sentence at the start setting up context (such as describing the issue being discussed) to give the reader some insight into what was being investigated.

14. Identify one area of statistics where your understanding could be improved.

I now understand how I can use the mean and standard deviation of data to make a sound, evidence backed, conclusion. I also now know how to tie this information to graphs to make an argument.

Evaluation Task Solutions

Solutions will vary between students depending on the topic they choose, and the variables they analyse. To create the best possible guide for educators, an example solution has been provided for each topic, along with feedback and a graded rubric.

Student Responses are in black, and teacher feedback is in red.

Blanding's Turtle Solution

Framing the Problem

1. How could the effectiveness of each strategy be measured?

For the effectiveness of each strategy, they can be measured by looking at trends in the estimated population. Other indicators that can be used for comparison are Nest success rate (higher is better because it means more hatchlings), Adult mortality rate (lower is better because adults are important for future breeding), Road mortality (for habitat expansion, lower is better because it could suggest fewer turtles are dying from vehicles), and stability and reliability of the population estimates (less variability is better because it means the population is more stable and the estimates are more consistent).

2. How could you compare the effectiveness of each strategy?

One way to compare the effectiveness is by looking at the performance for each from the first year (2012) to the most recent year (2022). Looking at the change in the variables over the time period above and comparing can show the effectiveness overall of the strategies. Comparison of trends in the positive factors mentioned above could be indicators for which method could be more effective depending on what you are wanting to analyze.

Your methods for indicating effectiveness and comparing them are clear. I would have liked a more in depth explanation into how the differing approaches led to your choice in comparison methods.

Understanding the Data

3. Identify the variables you will use to compare the two strategies. Explain why these variables are relevant. This question is no longer in our question set.

When comparing the two strategies I will use the following variables:

Estimated population: This is one of the main outcomes to compare as the goal is to protect and grow the Blanding's turtle population

Nest success rate: This variable is important because higher nest rates usually lead to more young turtles entering the population.

Adult mortality rate: Adult turtles are slow to replace, so lowering adult mortality is important for long term population survival.

Conservation cost: This matters because a strategy realistically needs to be cost effective to make sure it is sustainable for the long term so that it can be funded for a longer period of time.

Population variability: This would be a good variable to calculate to show how spread out or uncertain the population estimates are. Lower variability suggests more confidence and stability.

Variable selections are thoughtful and clearly justify how they contribute to evaluating the strategies effectiveness.

4. Calculate appropriate statistical measures (e.g., mean, median, range, IQR, standard deviation) for each strategy. Show one fully worked example for each type of calculation used.

Habitat Expansion (2012 to 2022, 11 years)

Estimated Population

Mean ≈ 452.1

Median = 455

Range = 475 − 420 = 55

IQR = 468 − 435 = 33

Standard deviation ≈ 17.9

Nest Success Rate (%)

Mean = 54.0

Median = 55

Range = 58 − 48 = 10

IQR = 56 − 52 = 4

Standard deviation ≈ 3.0

Adult Mortality Rate (%)

Mean ≈ 5.14

Median = 5.0

Range = 6.2 − 4.5 = 1.7

IQR = 5.6 − 4.7 = 0.9

Standard deviation ≈ 0.55

Cost ($k)

Mean ≈ 154.1

Median = 155

Range = 185 − 120 = 65
IQR = 175 − 130 = 45

Standard deviation ≈ 22.9

Habitat Connectivity (Connected habitats only, 6 data points)

Connected Estimated Population

Mean ≈ 441.8

Median = 444

Range = 465 − 415 = 50

IQR = 458 − 425 = 33

Standard deviation ≈ 19.4

Connected Nest Success Rate (%)

Mean ≈ 54.2
Median = 55
Range = 58 − 49 = 9
IQR = 57 − 51 = 6
Standard deviation ≈ 3.5

Connected Adult Mortality Rate (%)

Mean ≈ 5.08

Median = 5.0

Range = 5.8 − 4.6 = 1.2

IQR = 5.4 − 4.7 = 0.7

Standard deviation ≈ 0.45

Connected Cost ($k)

Mean = 165.0
Median = 167.5
Range = 195 − 130 = 65
IQR = 185 − 145 = 40
Standard deviation ≈ 24.7

Mean (Strategy A estimated population)
Data from 2012 to 2022

420, 428, 435, 448, 455, 452, 460, 468, 462, 470, 475

Adding values together: 4973

4973 / 11 = 452.091

Median (strategy A estimated population)

420, 428, 435, 448, 455, 452, 460, 468, 462, 470, 475

Finding the middle values (since there are eleven values).

Median is 455 (6th)

Range (Strategy A road mortality count)

Road mortality values: 41, 39, 36, 32, 29, 31, 28, 26, 30, 27, 25

Range = max - min

= 41 - 25

= 16

IQR (Interquartile Range) Strategy A estimated population

Sorted list: 420, 428, 435, 448, 452, 455, 460, 462, 468, 470, 475

Since there are an odd number of values we exclude the median. After excluding the median we then split the lower half (first 5): 420, 428, 435, 448, 452 then the upper 5: 460, 462, 468, 470, 475.

Q1 = median of lower half = 435

Q3 = median of upper half = 468

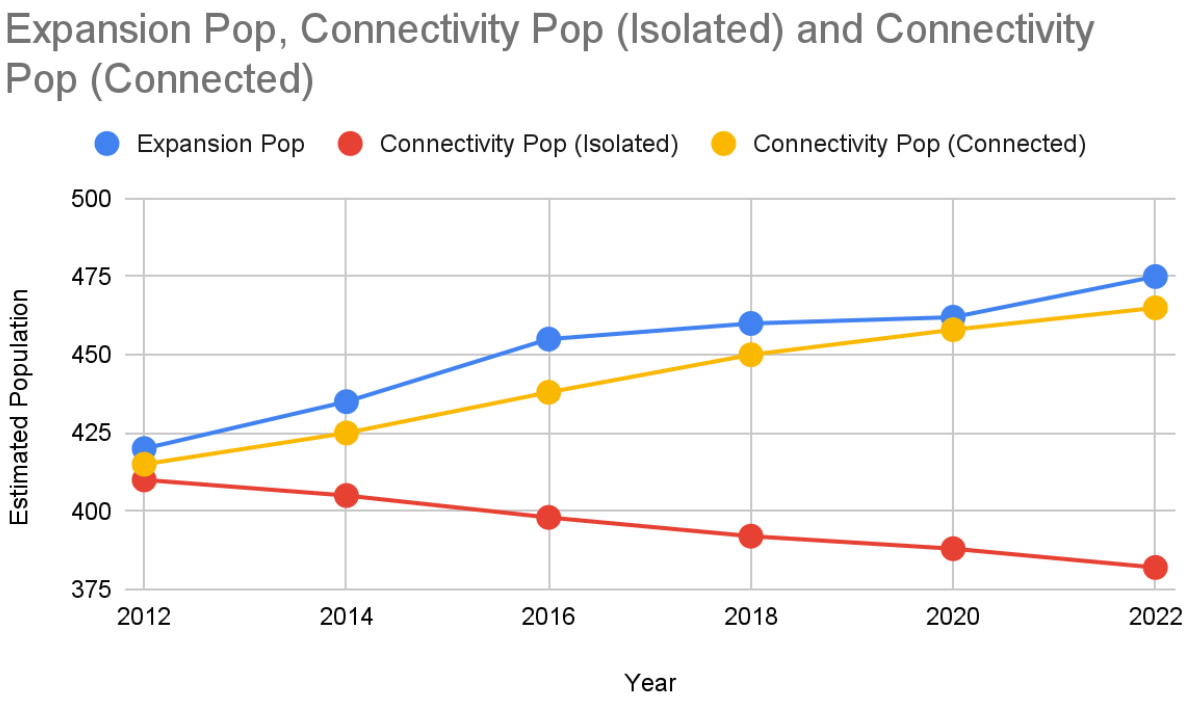
IQR = Q3 - Q1

= 468 - 435

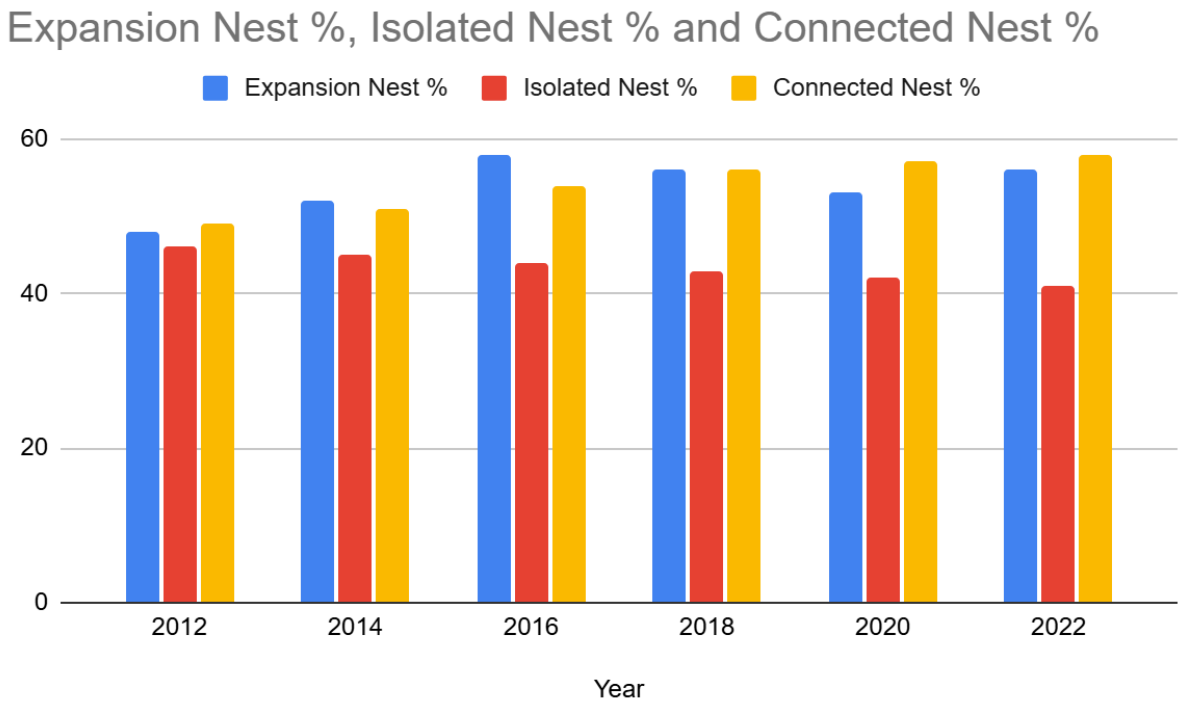
= 33

This data would be better displayed in a table.

5. Create at least two graphs that support a comparison of the strategies (scatter plot, bar graph, histogram, boxplot, or pie chart). Explain why each graph is appropriate.



For the multi line graph chart, I wanted to create a chart that compares estimated population over time for strategies. This graph shows values for the same years between the different methods and the lines show the trends for each method. This graph allows me to see how the population is affected with each method which can help inform me on the effectiveness of a restoration method. Population is important to take into consideration as a trend in population growth or decrease informs us the effectiveness of a method.



For this graph I decided to use a clustered column chart to compare nesting success rates for each strategy for the same years. Because nesting is a percentage the column chart makes it easier to directly compare each strategy year by year against each other. Nesting rate was selected because it is an important indicator of whether a restoration strategy is helping the population recover.

Graph choices make sense for the chosen variables and are clearly presented. Showing more factors leading to a decrease in Blanding's turtle population, such as temperature and road mortality, would help strengthen your argument. Remember to label your graph axis.

Interpreting the Evidence

6. Using statistical evidence, compare the short/long term outcomes of each strategy.

To compare short-term outcomes, I used the first shared time period where both strategies have data: 2012 to 2016. I compared Habitat Expansion to Habitat Connectivity (Connected) because "Connected" represents the restoration action of adding corridors. In the short term, Habitat Expansion shows larger improvements in the main outcome measures. Estimated population increased from 420 to 455 under Habitat Expansion, which is a change of +35 turtles (about +8.3%). Over the same period, Connectivity (Connected) increased from 415 to 438, which is +23 turtles (about +5.5%). Nest success rate also improved more under Habitat Expansion, rising from 48% to 58% (+10 percentage points) compared to Connectivity (Connected), which increased from 49% to 54% (+5 percentage points). Adult mortality rate decreased under both strategies, but the decrease was larger for Habitat Expansion: 6.2% to 4.9% (a drop of 1.3

percentage points) compared to Connectivity (Connected): 5.8% to 5.1% (a drop of 0.7 percentage points). The cost increase from 2012 to 2016 was the same for both strategies (+$30K), so Habitat Expansion produced bigger short-term gains for the same increase in cost.

Looking at the overall trend from 2012 to 2022, both strategies are linked to long-term improvement, but they show strengths in different areas. For estimated population, Habitat Expansion increases from 420 to 475 (a total change of +55 turtles), while Connectivity (Connected) increases from 415 to 465 (+50 turtles). This suggests both strategies support long-term population growth. However, the Connectivity data also includes an isolated condition, and that population decreases from 410 to 382 (−28 turtles), which suggests that leaving habitats isolated is not effective long term compared to connected habitats. For reproduction, Connectivity (Connected) shows a slightly stronger long-term pattern in nest success, increasing from 49% to 58% (+9 points), while Habitat Expansion increases from 48% to 56% (+8 points). For survival, Habitat Expansion shows a bigger long-term decrease in adult mortality, dropping from 6.2% to 4.5% (−1.7 points) compared to Connectivity (Connected), which drops from 5.8% to 4.6% (−1.2 points). Overall, the long-term trends suggest that both strategies can improve population outcomes over time, but Connectivity (Connected) appears slightly stronger for nest success, while Habitat Expansion shows stronger improvement in reducing adult mortality.

You clearly connected trends, variability and statistics for all the strategies. Your future projections would have been stronger if you included linear regression data to justify your arguments.

7. Identify any important trade-offs or real-world constraints (e.g., cost, variability, time, practicality) shown in the data.

One major trade-off is cost. In the shared years, Connectivity (Connected) is usually more expensive than Habitat Expansion. For example, in 2022, Habitat Expansion costs $185K while Connectivity (Connected) costs $195K. This matters because even if Connectivity gives strong nesting outcomes, it may be harder to choose if funding is limited. Another trade-off is stability and variability. The Connectivity dataset includes population variability (±), which shows how consistent the population is. Over time, Connected variability decreases from ±32 to ±24, suggesting the population becomes more stable. In contrast, Isolated variability increases from ±48 to ±62, suggesting the population becomes less stable. This is important because a stable population is usually more sustainable long term, even if the average population is similar. There are also practical constraints related to time and feasibility. Habitat Expansion requires increasing protected habitat area (from 180 km² to 240 km²), which can be slow because it may involve land protection policies, costs, and agreements. Connectivity requires building and maintaining corridors (corridor length increases from 12 km to 45 km), which may be limited by roads, private land, and where corridors can realistically be placed. Finally, the strategies reduce risk in different ways: Habitat Expansion is linked to a large decrease in road mortality (from 41

to 25), while Connectivity is linked to improving nest success and stabilizing the population. This means decision-makers have to balance what outcome is most important and what is most realistic to implement.

Discussing the Data

8. Discuss at least two strengths and two limitations of the data used in your comparison.

One strength of the data is that it includes multiple variables, not just population. I can compare strategies using nest success, adult mortality, cost, and survey effort, which gives a more complete picture of restoration effectiveness. A second strength is that the dataset covers multiple years, which allows me to look at trends and patterns over time instead of making a conclusion based on only one year. A limitation is that the two strategies do not have the same number of data points. Habitat Expansion has yearly data from 2012 to 2022, while Connectivity only provides data every two years, which makes comparisons less fair and reduces confidence in long-term conclusions for Connectivity. Another limitation is that this is not a controlled experiment, so other factors like weather, changes in predator populations, habitat disturbance, or road traffic (human related changes) could also affect the results. Because those factors are not controlled for, I cannot prove the strategy alone caused the outcomes. I would need to look into other variables and how those would be taken into account for the bigger picture.

Including a detailed explanation about the short term variability due to the un-controlled experiment is a good point. I would like you to elaborate on how that would affect the long term outcomes of the experiment. Would these factors average out, and become irrelevant, or would they cause a large skew in the results?

9. Based on the data, determine which strategy appears to be more effective or sustainable. Support your decision with statistical evidence.

Based on the data, Habitat Expansion appears to be the more effective and sustainable strategy overall, even though Habitat Connectivity (Connected) performs slightly better in one area. Over the time period, Habitat Expansion shows stronger long-term improvement in the main outcome we care about, which is the estimated population. The estimated population increases from 420 in 2012 to 475 in 2022, which is a difference of 55 turtles. Connectivity (Connected) also improves, increasing from 415 to 465, a gain of 50 turtles, but the increase is slightly smaller than Expansion. When I look at reproduction, Connectivity (Connected) has a small advantage because nest success rises from 49% to 58% (an increase of 9 %) compared to Habitat Expansion, which rises from 48% to 56% (an increase of 8 %). However, Habitat Expansion shows a stronger improvement in adult survival, which is important for turtles because adults take a long time to grow up and replace. Adult mortality drops from 6.2% to 4.5% under Expansion (a decrease of 1.7 %), while Connectivity (Connected) drops from 5.8% to 4.6% (a decrease of 1.2 %). Cost is also an important sustainability factor. Both strategies increase by $65K over the time period, but in comparable years Connectivity is generally more expensive,

such as in 2022 where Expansion costs $185K and Connected costs $195K. Finally, Habitat Expansion includes a major real-world benefit that supports long-term sustainability: road mortality decreases from 41 to 25, which is 16 fewer road deaths, and reducing adult road deaths can have a direct impact on keeping the breeding population stable. Overall, even though Connectivity (Connected) slightly improves nest success more, Habitat Expansion provides the strongest combination of population growth, improved adult survival, and lower road mortality, while also being slightly cheaper in the same years, so it appears more effective and sustainable based on this dataset.

In reality, the cost of implementing these types of strategies is a major factor. In the future, try elaborating on the costs, indicate whether the costs will be ongoing, or temporary, and how this could shape the choice in strategies. The use of linear regression to strengthen your future projections would be beneficial.

10. Suggest at least one follow-up investigation or additional data that could improve future decision-making.

One follow-up investigation that would improve decision-making is collecting yearly data for the Connectivity strategy, instead of only every two years, so the trend can be compared more fairly to Habitat Expansion. I would also want additional data on juvenile survival (how many hatchlings survive to become adults), because nest success alone does not guarantee long-term population growth if young turtles do not survive.

These are all strong follow up points. To make them even better you could indicate the questions these investigations would answer, and address any limitations associated with them.

Reflection and Communication

11. What main points will be essential to convey your message? Use these points to build your presentation.

The main points I need to communicate in my presentation are that I compared two restoration strategies for Blanding's turtle using the same outcome variables and the same years where possible, and then used statistics and graphs to decide which strategy is more effective and sustainable. I need to clearly explain what each graph shows, especially the overall trends. The population trend graph matters because it shows whether the strategies are linked to population growth over time, and it lets the audience see which strategy increases faster in the shared years. The nest success graph matters because it shows whether reproduction is improving and whether one strategy consistently supports higher success rates. I also need to explain why adult survival is a key factor for turtles, since adult turtles are slow to replace, so a strategy that lowers adult mortality and road deaths can have a big long term impact. Finally, I need to connect the evidence to a recommendation by comparing not just which strategy performs slightly better in

one category, but which strategy gives the best overall combination of population growth, survival, and cost in the real world.

Link to presentation for turtle: Presentation Link (google drive)

Your points are thoroughly planned and thought out.

Most findings are explained in a way that all math skill levels could understand. When it comes to your tables, I would suggest including a description of what the data shows, or its relevance in that particular format.

The information on your slides are clearly stated and they follow a particular style, but they're also very busy with text, taking away from the viewer's engagement. Next time consider reducing the amount of text, and instead include a script that would accompany the slides.

12. After completing your presentation of findings, reflect on this task and describe one thing that went well, one thing that challenged you, and one thing you would have done differently if given the option to try again.

One thing that went well with the task was being able to create the graphs to represent the data in an effective manner. The graphs showed clear trends which made it easier to understand the outcomes of the strategies over the given compared time periods. One thing that challenged me was selecting which variables to investigate and their effect on one another. I wanted to ensure that I was making meaningful conclusions from the dataset which were insightful into which strategy was better. One thing I would have done differently if given the option to try again would be how I presented the information. If I was given the option I think I would want to do a poster so that all the information is displayed on the same page making it easier to jump around to show different parts of my analysis.

Criteria	Level 1 (50–59%)	Level 2 (60–69%)	Level 3 (70–79%)	Level 4 (80–100%)
Understanding of Restoration Strategies (Knowledge and Understanding, Application) **7/10 marks**	Demonstrates a limited understanding of the two strategies. Purpose and goals are unclear or inaccurate.	Demonstrates a partial understanding of both strategies, with some minor misunderstandings.	Demonstrates a clear and accurate understanding of both strategies and what each aims to improve.	Demonstrates a thorough understanding of both strategies, including how and why they differ in approach and goals.
Selection and Use of Data (Thinking) **5/5 marks**	Uses very little relevant data or relies on inappropriate relationships. Data selection is not justified.	Uses some relevant data, but selection may be incomplete or weakly justified.	Selects appropriate variables and explains why they are relevant to comparing strategies.	Thoughtfully selects multiple relevant variables and clearly justifies how each contributes to evaluating strategy effectiveness.
Statistical Analysis (Knowledge and Understanding) **11/15 marks**	Statistical measures are missing, incorrect, or inappropriate. Graphs are unclear or not meaningful.	Some appropriate statistical measures and graphs are used, but with errors or limited explanation.	Appropriate statistical measures and graphs are correctly calculated and clearly presented.	Statistical measures and graphs are accurate, well-chosen, and clearly linked to the research question, showing strong analytical judgment.
Interpretation of Evidence (Application) **11/15 marks**	Interpretations are vague, inaccurate, or not supported by data.	Interpretations reference data but may be superficial or partially unsupported.	Interpretations clearly connect trends, variability, and statistics to the comparison of strategies.	Provides insightful interpretation of trends, variability, and projections, demonstrating deep understanding of what the data shows and implies.

Justification of Chosen Strategy (Application) **13/15 marks**	A strategy is chosen with little or no supporting evidence.	A strategy is chosen with some evidence, but the argument lacks clarity or depth.	A clear strategy is justified using relevant statistical evidence and comparisons.	A compelling, well-structured argument clearly supports one strategy, addressing both short- and long-term outcomes and trade-offs.
Evaluation of Data Strengths and Limitations (Application) **4/5 marks**	Little or no discussion of data strengths or limitations.	Identifies some limitations or strengths, but discussion is brief or unclear.	Clearly discusses strengths, limitations, and variability, and explains how these affect confidence in conclusions.	Thoughtfully evaluates data quality, bias, variability, and uncertainty, showing strong critical thinking.
Follow-Up Investigation (Application) **3.5/5 marks**	Follow-up suggestions are missing or unrealistic.	Follow-up investigation is suggested but lacks detail or relevance.	Suggests a reasonable and relevant follow-up investigation or additional data.	Proposes a well-reasoned follow-up investigation that clearly addresses limitations or unanswered questions.
Communication and Presentation (Communication) **14/20 marks**	Communication is unclear, or difficult to follow. Presentation is disorganized or poorly suited to the chosen format.	Some effort to communicate clearly.. Organization is inconsistent or format is only partially effective.	Information is clearly organized, the format is appropriate, and information is explained in a way that all math skill levels can understand.	Communication is exceptionally clear, engaging, and accessible. Presentation is polished, well-structured, and makes excellent use of the chosen format.

Total: 68.5/90= 76%

Emerald Ash Borer Solution

Framing the Problem

1. How could the effectiveness of each strategy be measured?

Removal and Replacement:

The infestation rates and forest composition can be compared for both before and after removal and planting efforts.

Containment Zones:

The infestation levels in treated zones versus untreated zones can be compared over time.

2. How could you compare the effectiveness of each strategy?

The percentage of total number of trees, number of infested trees, and canopy loss can be compared for both strategies over a similar time period. The cost can also be compared.

Good job identifying the variables you are going to analyse to measure the effectiveness of each strategy. When it comes to picking those specific variables to look at for the comparison, what made you pick those ones specifically?

Understanding the Data

3. Identify the variables you will use to compare the two strategies. Explain why these variables are relevant.

Percentage in change will be used to compare the change in data over the recorded time period.

The standard deviation will be used to analyse the cost variability for each strategy.

Statistical Analysis

4. Calculate appropriate statistical measures (e.g., mean, median, range, IQR, standard deviation) for each strategy. Show **one** fully worked example for each type of calculation used.

	Removal and Replacement	**Treated**	**Untreated**
Ash Tree Loss (%)	-83	-60	62
New Infected Trees (%)	-93	49	64
Mean Cost/Year ($/yr)	1925	904	0
Cost/Year SD ($/yr)	1203	610	0

Sample Calculations for Treated Zone

Percentage (Ash trees lost change)

% change = -100% - 100%(2014 value - 2022 value)

= -100% - 100%(2000 - 5000)

= -60%

Percentage (Newly Infected Trees Change)

% change = 100% - 100%(2014 value - 2022 value)

= 100% - 100%(7000 - 13600)

= 49%

mean cost

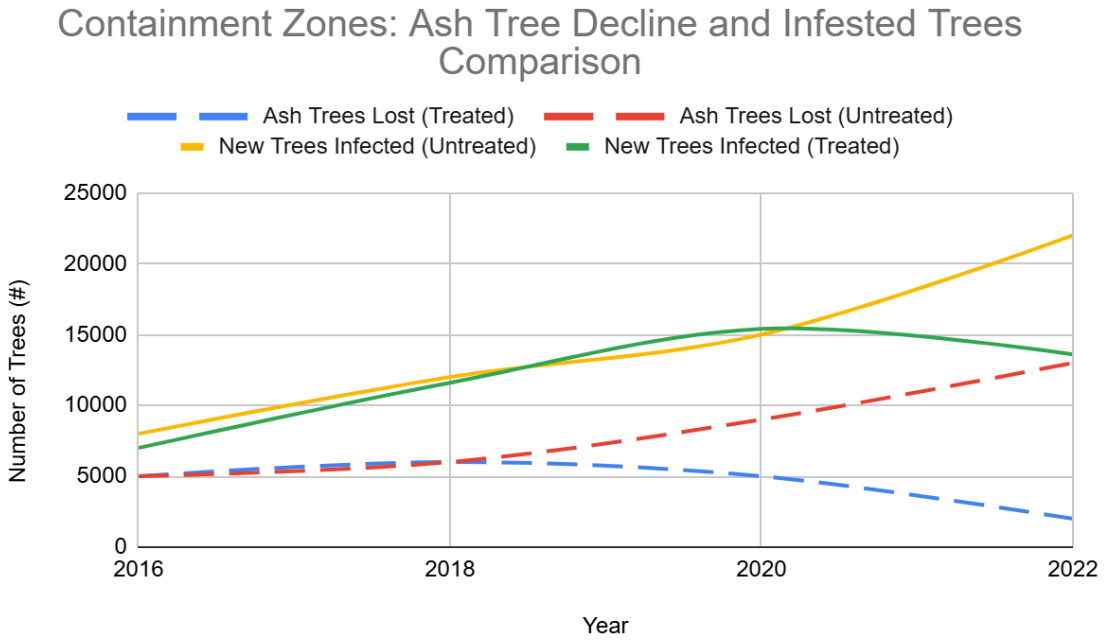
$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i = \frac{1}{5}\sum_{i=1}^{5} x_i = 904$$

Standard Deviation for Cost

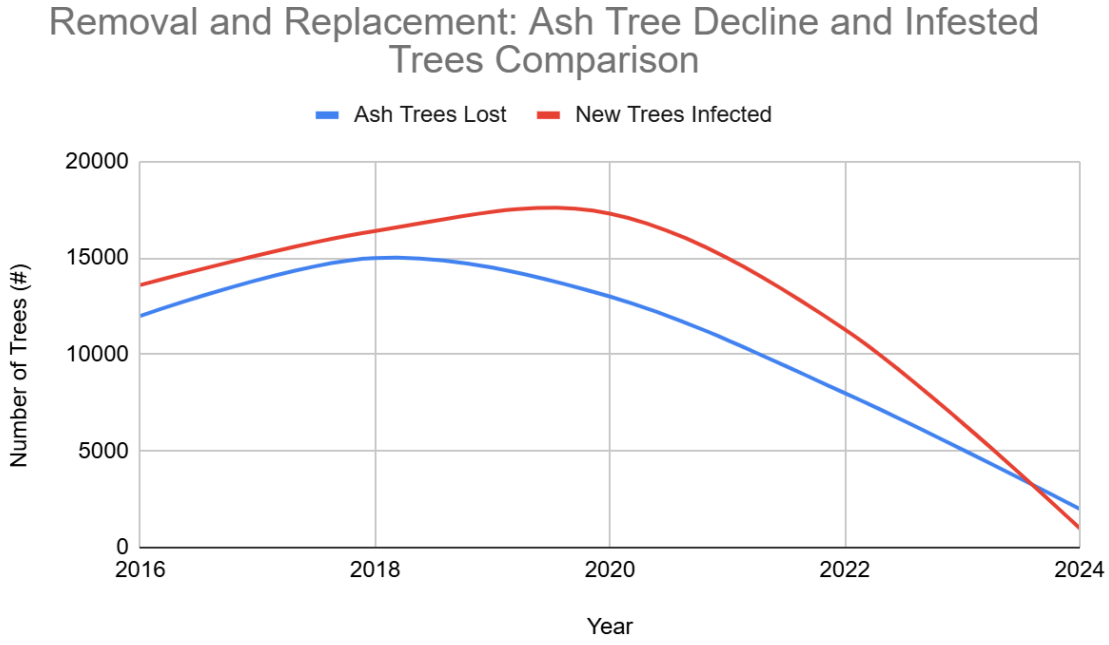
$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}} = \sqrt{\frac{\sum (x_i - 904)^2}{5}} = 610$$

The table allows for an organized arrangement of the data that is easy to read. For the calculations I like how you performed the calculations for different variables.

5. Create at least two graphs that support a comparison of the strategies (scatter plot, bar graph, histogram, boxplot, or pie chart). Explain why each graph is appropriate.

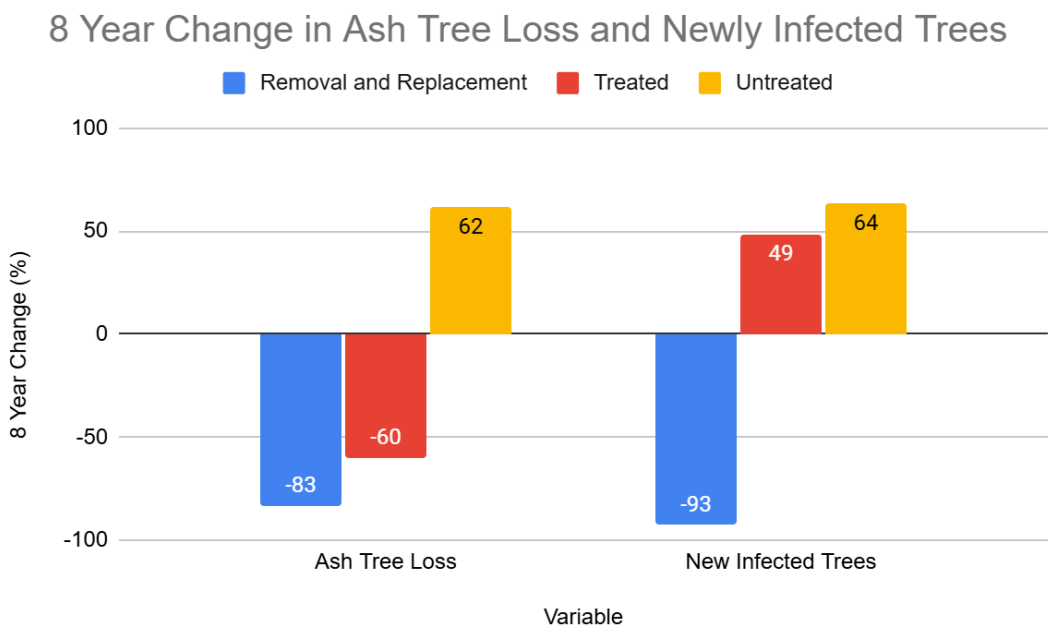


The above graph shows the decline in ash tree populations for both the treated and non-treated groups for the containment zones strategy. The graph also shows the new infested trees for both groups.



The graph shows the same variables as the previous graph, but this time for the removal and replacement strategy.

Comparing the initial and final points as percentage of change for each graph's *Ash Trees Lost* portion will show which strategy saves the most trees.



This bar graph shows how well the different strategies work for reducing the loss of ash trees and newly infected sites. Over 8 years, the removal and replacement strategy and the treatment strategy both experience a decrease in the amount of trees lost each year. The removal and replacement strategy also shows a huge reduction in the number of newly infected trees (-93%), whereas the other methods continued to experience an increase in the number of newly infected trees.

There is a good variety of graphs here to represent the data in different ways. The graphs axes are correctly labeled and brief explanations are provided. In the explanations for each graph, what are some trends that you notice? How do these graphs specifically help represent the data visually?

Interpreting the Evidence

6. Using statistical evidence, compare the short/long term outcomes of each strategy.

The untreated trees in the containment zones suggest that, as time goes on, the number of infestations, and the number of trees lost, will continue to increase exponentially until there are no ash trees remaining.

The treated trees in the containment zones suggest that, as time goes on, the number of infestations will tend towards zero, suggesting that trees can recover from the infestation. It also suggests that the number of ash trees lost will go to zero.

The removal and replacement strategy shows that, as time goes on, the number of infested trees will decrease to zero, and so will the number of trees lost.

7. Identify any important trade-offs or real-world constraints (e.g., cost, variability, time, practicality) shown in the data.

The treatment of trees for the containment zone strategy may cause unintended effects within the ecosystem, as the treatment may be toxic to more than just the emerald ash borer, and will likely enter the water system.

The removal and replacement strategy will result in a large amount of young trees, which may impact the ecosystem.

Discussing the Data

8. Discuss at least two strengths and two limitations of the data used in your comparison.

The datasets both allow for the comparison of ash tree decline and the number infested trees over time. This makes for an easy comparison between the two strategies.

Both datasets are taken over a similar time period, ensuring that they're accounting for the same effects of climate change and other time period related external factors.

The datasets don't show the regions the data was collected from. If the trees are in entirely different areas, this could skew the results due to external local factors.

Each strategy only has data for 10 years, resulting in a very small amount of data. Having more data would allow for more conclusive results.

For the treated containment zones dataset, it would be helpful to get a measure of other plants in the area. This would help determine whether the treatment is affecting other parts of the ecosystem.

9. Based on the data, determine which strategy appears to be more effective or sustainable. Support your decision with statistical evidence.

The treated containment zones experienced a 60% decrease in ash tree loss, while the untreated containment zone experienced a 62% increase in ash tree loss. The treated ash tree zone showed to have a substantial impact on preventing the death of ash trees. In contrast, the removal and replacement method produced an 83% decrease in ash tree loss, making it the most effective method for the prevention of ash tree death.

In terms of newly infected trees, the untreated containment zones experienced a 64% increase over 8 years, whereas the treated zones experienced a 49% increase over the same time period. It should be noted that, in the final year of data collection, the number of newly infected trees

began to decline (12% reduction compared to the previous year), indicating that this number will likely reach zero within the next few years. The removal and replacement strategy experienced a 93% reduction over a similar time period (10 years) in newly infected trees, indicating that it is far more effective when it comes to decreasing the amount of newly infected trees.

The containment zones treatment strategy cost, on average, $904 per year with a standard deviation of $610, the untreated zones cost $0 per year with a standard deviation of $0, and the removal and replacement strategies cost $1925 per year with a standard deviation of $1203.

Although the removal and replacement strategy costs marginally more than the treatment strategy ($1021 per year), its effectiveness at reducing the number of ash tree loss makes up for the difference. Since the removal and replacement method is so much more effective, the emerald ash borer will be eradicated over a shorter period of time, and therefore this additional cost will be paid for a short period of time, potentially becoming a more cost effective strategy.

This is very insightful, I like how you talked about both methods and compared both before stating which strategy you think is better. Additionally a good use of data from the investigation incorporated into your response to strengthen your point.

10. Suggest at least one follow-up investigation or additional data that could improve future decision-making.

A follow up study pertaining to the effects of the treatment in containment zones on the local environment would affect the viability of the strategy. If the treatment has no, or limited, effects, the strategy can be a viable option. In contrast, if the treatment harms the surrounding environment, this strategy would no longer be reasonable.

For the removal and replacement strategy, a large amount of the ash tree population will be young. A study on the amount of surviving young ash trees for the removal and replacement strategy would help indicate whether the strategy is effective in the long run, or if the ash tree population would result in a dangerous position due to the strategy.

I like how for this question you provided a follow-up investigation for both of the strategies. For these follow-up investigations I am wondering if you could provide some references? Additionally is there any data from the provided dataset that could help provide you with an area to investigate (what prompted you to suggest these follow-up investigations specifically?).

Reflection and Communication

11. What main points will be essential to convey your message? Use these points to build your presentation.

I will include a brief overview of each strategy.

I will argue that the removal and replacement strategy is best using the data displayed in the table from question 4, the graphs from question 5, and the discussion points from question 9.

I will review the strengths and limitations of the data from questions 7 and 8.

I will then suggest follow up studies as was discussed in question 10.

12. After completing your presentation of findings, reflect on this task and describe one thing that went well, one thing that challenged you, and one thing you would have done differently if given the option to try again.

Deciding which data was most important was the most challenging part. There was a lot of data that I chose not to include because I felt the points were best made with the statistics that I chose.

Deciding which data to include on the poster was the most challenging part. I had to decide which data was most important to get a point across, as not all of it would fit.

The data clearly showed that one strategy was better than the other, if the differences weren't as extreme, it would have been difficult to make a sound argument for which strategy is better.

If I were to do the project again, I would have used linear regression to determine how long it would take each strategy to eradicate the emerald ash borer, so the cost for each strategy could be accurately calculated.

Saving The Ash Trees

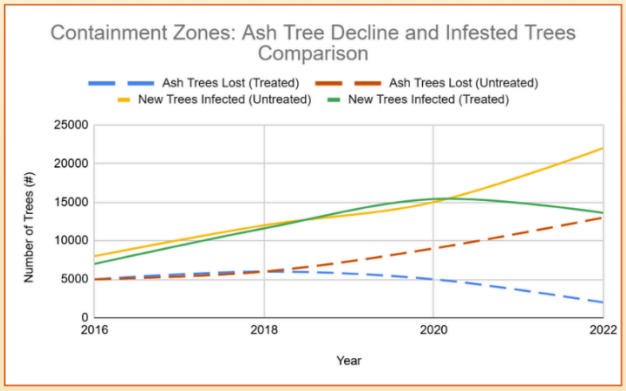
A comparison of strategies for rescuing ash trees from the invasive emerald ash borer

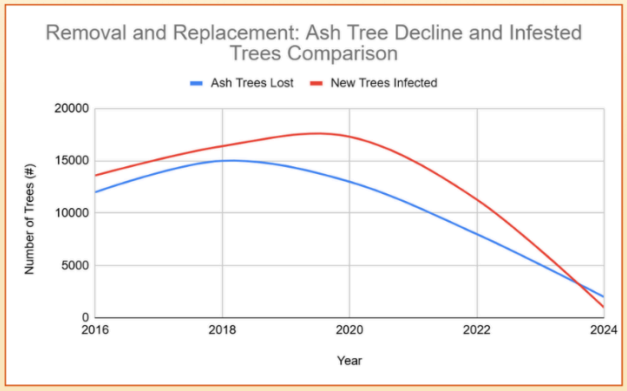
Removal and Replacment

Infected trees are removed and replaced with a variety of native species

Containment Zones

Infected areas are contained. Some areas are treated with insecticides, whereas others are not.





Untreated Containment Zones

- 62% increase in ash tree loss
- 49% increase in newly infected zones over 8 years
- $0±$0per year

Treated Containment Zones

- 60% decrease in ash tree loss
- 64% increase in newly infected zones over 8 years
- $904±$610 per year

Removal and Replacement

- 83% decrease in ash tree loss
- 93% decrease in newly infected zones over 8 years
- $1925±$1203per year

Study Strengths

- Data collected across both studies were done in a way that makes them comparable.
- Data was taken over a similar time periods, ensuring that the effects of climate change were equally accounted for.

Study Limitations

- Studies may have occurred in different regions
- Data was only available for up to 10 years.
- Dataset excluded local plant life reactions to treatment strategies.

Conclusion

Although the removal and replacement strategy costs marginally more than the treatment strategy, its effectiveness at reducing the number of ash tree loss and new infections makes up for the difference. Since the removal and replacement method is so much more effective, the emerald ash borer will be eradicated over a shorter period of time, and therefore the additional costs will be paid over a short period of time, also becoming the more cost effective strategy.

Criteria	Level 1 (50–59%)	Level 2 (60–69%)	Level 3 (70–79%)	Level 4 (80–100%)
Understanding of Restoration Strategies (Knowledge and Understanding, Application) **10 marks** **7.5/10**	Demonstrates a limited understanding of the two strategies. Purpose and goals are unclear or inaccurate.	Demonstrates a partial understanding of both strategies, with some minor misunderstandings.	Demonstrates a clear and accurate understanding of both strategies and what each aims to improve.	Demonstrates a thorough understanding of both strategies, including how and why they differ in approach and goals.
Selection and Use of Data (Thinking) **5 marks** **3.5/5**	Uses very little relevant data or relies on inappropriate relationships. Data selection is not justified.	Uses some relevant data, but selection may be incomplete or weakly justified.	Selects appropriate variables and explains why they are relevant to comparing strategies.	Thoughtfully selects multiple relevant variables and clearly justifies how each contributes to evaluating strategy effectiveness.
Statistical Analysis (Knowledge and Understanding) **15 marks** **10.5/15**	Statistical measures are missing, incorrect, or inappropriate. Graphs are unclear or not meaningful.	Some appropriate statistical measures and graphs are used, but with errors or limited explanation.	Appropriate statistical measures and graphs are correctly calculated and clearly presented.	Statistical measures and graphs are accurate, well-chosen, and clearly linked to the research question, showing strong analytical judgment.
Interpretation of Evidence (Application) **15 marks** **13/15**	Interpretations are vague, inaccurate, or not supported by data.	Interpretations reference data but may be superficial or partially unsupported.	Interpretations clearly connect trends, variability, and statistics to the comparison of strategies.	Provides insightful interpretation of trends, variability, and projections, demonstrating deep understanding of what the data shows and implies.

Justification of Chosen Strategy (Application) **15 marks** **11/15**	A strategy is chosen with little or no supporting evidence.	A strategy is chosen with some evidence, but the argument lacks clarity or depth.	A clear strategy is justified using relevant statistical evidence and comparisons.	A compelling, well-structured argument clearly supports one strategy, addressing both short- and long-term outcomes and trade-offs.
Evaluation of Data Strengths and Limitations (Application) **5 marks** **3.5/5**	Little or no discussion of data strengths or limitations.	Identifies some limitations or strengths, but discussion is brief or unclear.	Clearly discusses strengths, limitations, and variability, and explains how these affect confidence in conclusions.	Thoughtfully evaluates data quality, bias, variability, and uncertainty, showing strong critical thinking.
Follow-Up Investigation (Application) **5 marks** **5/5**	Follow-up suggestions are missing or unrealistic.	Follow-up investigation is suggested but lacks detail or relevance.	Suggests a reasonable and relevant follow-up investigation or additional data.	Proposes a well-reasoned follow-up investigation that clearly addresses limitations or unanswered questions.
Communication and Presentation (Communication) **20 marks** **15.5/20**	Communication is unclear, or difficult to follow. Presentation is disorganized or poorly suited to the chosen format.	Some effort to communicate clearly.. Organization is inconsistent or format is only partially effective.	Information is clearly organized, the format is appropriate, and information is explained in a way that all math skill levels can understand.	Communication is exceptionally clear, engaging, and accessible. Presentation is polished, well-structured, and makes excellent use of the chosen format.

Total: 69.5 / 90 = 77%

Rough Work

Expectations

Learning Task

Overall Expectation

- D1. Analyse, interpret, and draw conclusions from one-variable data using numerical and graphical summaries;
- D2. Analyse, interpret, and draw conclusions from two-variable data using numerical, graphical, and algebraic summaries

Specific Expectation

- C2.5 collects data from primary or secondary sources and organize the data to answer a question or solve a problem.
- D1.1 Determine the relevant numerical summaries for one-variable data (mean, median, mode, range, interquartile range, variance, and standard deviation).
- D1.5 Interpret numerical and graphical summaries to describe the characteristics of a data set and explain what the data shows.

Evaluation Task

Overall Expectation

- E1. Design and carry out a culminating investigation that requires the integration and application of the knowledge and skills related to the expectations of this course.
- E2.Communicate the findings of a culminating investigation

Specific Expectation

- E 1.5 draw conclusions from the analysis of the data (e.g., determine whether the analysis solves the problem), evaluate the strength of the evidence (e.g., by considering factors such as sample size or bias, or the number of times a game is played), specify any limitations of the conclusions, and suggest follow-up problems or investigations

Summary

This culminating activity for MDM4U is designed to integrate statistical analysis with real-world, land-based environmental issues in Ontario. Through a connected learning task and

evaluation task, students apply data analysis skills to highlight a land-based problem and then use evidence-based reasoning to justify a potential solution. The structure emphasizes interpretation, communication, and informed decision-making using data.

Summary Feedback

I like how you're looking at both invasive species and species at risk. You could also have students make explicit connections between the two, how invasive species can cause native species to become at risk. I think this could become a very good LT/ET pair.

Learning Task: Understanding a Land-Based Problem

In the learning task, students select one of three Ontario-based environmental issues: a species at risk (species TBD), the spread of the Emerald Ash Borer as an invasive species, or pollinator loss. Using the provided hypothetical data, students investigate why their chosen issue is a problem. The focus is on interpreting data rather than collecting it, allowing students to concentrate on statistical reasoning and communication.

Students organize and analyze the data using methods they find appropriate. This may include measures such as mean, median, range, interquartile range, variance, and standard deviation. Students interpret data and explain what the data reveals about the environmental issue. Findings are communicated clearly through a report, presentation, or video to demonstrate why the selected issue warrants attention.

This task addresses Overall Expectations D1 and D2, with a focus on collecting and organizing data, determining relevant numerical summaries, and interpreting statistical evidence to draw meaningful conclusions.

Evaluation Task: Justifying a Restoration Strategy

Building directly on the learning task, the evaluation task requires students to propose and justify a restoration strategy for their chosen environmental issue. For each land-based problem, students are provided with two possible restoration strategies and accompanying data. Students must clearly identify what each strategy aims to improve and how success would be measured using statistical indicators.

Students compare the two strategies using data, considering short and long term outcomes, trade-offs, and real-world constraints. This includes analyzing trends, variability, and projected future outcomes to determine which strategy is more effective or sustainable. Statistical evidence, including summaries and graphs, must be used to support conclusions. Students are also expected to evaluate the strengths and limitations of the data, considering factors such as variability, potential bias, or sample size, and to suggest possible follow-up investigations.

Students communicate their findings through an approved format such as a written report, presentation, video, or poster.

This task fulfills Overall Expectations E1 and E2, emphasizing integrated application of course knowledge and clear communication of conclusions drawn from data.

Land-Based Problem Contexts

Each problem provides a distinct context for statistical investigation. Species at risk scenarios focus on declining populations due to habitat fragmentation and reduced breeding success, with restoration strategies centred on habitat expansion or improved connectivity. The Emerald Ash Borer case examines invasive species spread and tree loss, comparing removal and replacement strategies with containment zones. Pollinator loss investigations address declining pollinating plant counts and habitat limitations, with strategies including increasing plant diversity or expanding habitat through managed hives.

Learning Task

Must sees:

- Organize data to answer a question
- Calculate the relevant one-variable data (mean, median, mode, range, interquartile range, variance, and standard deviation).
- Interpret numerical and graphical data, and explain what it shows.

Land based problem choice: _____

Framing the Problem

1. Briefly describe the issue and explain why it is relevant to Ontario's land and ecosystems.
2. Write at least two clear, measurable statistical questions that can be answered using the provided data.

Understanding the Data

3. Identify at least three important variables you will focus on and explain why each is useful for understanding the problem. (mean, median, range, variance, standard deviation)
4. Choose and justify at least two ways to organize or display the data (e.g., table, histogram, box-and-whisker plot, scatterplot). Explain why each representation is appropriate for your variables.

Numerical Analysis

5. Calculate the variables you identified in question 3. Show one worked example for each type of calculation you use.
6. Show your calculated data in a clear table. Label all columns and include units where appropriate.
7. Create at least two different graphs that support your argument. Explain why each graph is appropriate.

Interpreting the Statistical Evidence

8. How does the deviation in your data affect how confident you are in your conclusions?
9. Identify any trends, clusters, or outliers. Explain what they suggest about your topic.
10. What claims can you make about past, present, and future conditions based on your data? Clearly state any assumptions you make.

Evaluating the Data

11. Discuss at least two strengths and two limitations of the dataset, and how they might affect your conclusions. Consider sample size, time span, variables included, or missing information.
12. Explain what additional data or changes would improve the reliability of your results.

Reflection and Communication

13. What are the possible implications of your conclusions?If a decision maker acted on your findings, how could the land, species, or community change? Explain at least one potential positive outcome, and possible unintended consequence.
14. Reflect on what the data revealed that a simple description of the issue would not.
15. How will you present your findings so that a non-expert can understand your conclusions and evidence? Justify your choice of format (e.g., report, poster, video, presentation, etc.).

Evaluation Task

Must sees:

- Determine whether the analysis solves the problem
- Evaluate the strength of the evidence (sample size or bias)
- Limitations in conclusions
- Follow up problems/investigations

Question ideas:

Instructions:

- analyze possible restoration strategies for your chosen land-based problem using the provided datasets.

- Task is to compare strategies, justify a recommendation, evaluate the evidence, and communicate your findings in an appropriate format (report, presentation, poster, video, etc.).

1. For your restoration strategy, clearly identify:
    a. What specific aspect of the problem it aims to improve (e.g., population size, infestation rate, pollinator abundance), and
    b. which two or more quantitative indicators from the dataset you will use to measure whether the strategy is successful (e.g., mean, median, rate of change, standard deviation)
2. For each of the restoration strategies:
    a. Calculate at least three relevant numerical summaries (such as mean, interquartile range, standard deviation, or percentage change over time).
    b. Present your results in a clearly labeled table, including units where appropriate
3. Create at least two different graphs that allow you to compare the effectiveness, variability, or trends of the two strategies (for example, side-by-side box plots, overlaid line graphs, or grouped bar charts). For each graph explain:
    a. Why you chose that type of graph, and
    b. What is reveals about the strategies
4. Based on your statistical analysis and graphs, which strategy do you recommend? In your justification, address:
    a. short-term outcomes (immediate effects on the problem),
    b. Long-term outcomes (projected trends or sustainability),
    c. At least one trade-off or constraint (e.g., cost, practicality, risk to other species, variability in results), and
    d. How confidence you are in your recommendation given the data quality
5. Evaluate the quality of your evidence by discussing:
    a. At least two strengths of the dataset (e.g., sample size, time span, consistency),
    b. at least two limitations (e.g., missing variables, possible bias, short time frame, lack of control groups), and
    c. how these strengths and limitations affect the reliability of your conclusions
6. Does your analysis fully solve the original problems you studies in the learning task, or does it only partially address it? Explain using numerical and graphical evidence, and identify at least one limitation or gap in the data that prevents you from making a stronger conclusions.
7. Suggest two specific follow-up investigations or additional data that would help you make a more confident, evidence-based decision about which restoration strategy to implement. For each suggestion, explain:
    a. What new data you would collect or what relationship you would investigate, and
    b. How it would strengthen your conclusions or address a limitation you identified

8. Imagine you are presenting to a non-expert decision maker (e.g., a community council, conservation group, or local landowner).
    a. What are key statistical findings you would highlight to convince them your recommended strategy is necessary?
    b. How will you present the evidence so it is clear, persuasive, and accessible to someone without a statistics background?
    c. Describe at least one potential positive impact your strategy could have on the land, specifics, or community, and one possible unintended consequence that should be monitored.

Presentation format

Present your findings through a written report, presentation, poster, video, or another method approved by your teacher. Your presentation must include your statistical summaries, graphs, justification, evaluation of evidence, and recommendation.

Based on the data collected how big is your problem

What are some other factors that you must take into account when analyzing these variables together? What are some considerations when looking at this data

What are some other relationships you can explore between your problem and other factors you think would be meaningful

If you were an environmentalist looking to convince the world that this is a problem, how and what would you use to show this.

Based on the data what are some short and long term outcomes what do you think is happening and affecting species

Evaluation Task Rubric

- Organizes data in a way that makes sense
- Chooses to calculate relevant data
- Calculations are correct
- Deviations in data suggest a strong argument
- Calculations and graphs justify the argument
- Data interpretations make sense
- Findings are presented clearly and accurately

Methodology

OpenAI's ChatGPT (Feb. 2, 2026 version) was used to generate the hypothetical data used for both the learning task and evaluation task datasets. Below are the prompts used to generate the data tables.

Learning Task Dataset Prompt

> Below is a learning task for a grade 12 data management class. Generate hypothetical data that can be used for each problem of the given task. Include years 2010 to 2016, different regions with some that can be used as a control (ash borer), population data, habitat loss, temperature, costs, and any other variables that will be relevant. Include variability in the data.
>
> Land-Based Problem Options
>
> Review and choose one of the following land-based problems to tackle for this assignment.
>
> Option 1: Blanding's Turtle
>
> Blanding's turtle is an endangered species in Ontario that depends on wetland habitats and surrounding land to survive. The data focuses on population and habitat-related measures over time, allowing you to analyze trends and variability. This option includes fewer data points, which may make it easier to manage, but conclusions may be less certain.
>
> Option 2: Emerald Ash Borer
>
> The emerald ash borer is an invasive insect that has spread through parts of Ontario, affecting ash tree populations in both urban and natural areas. The data tracks changes over time and across multiple regions, allowing you to examine patterns of spread, impact, and monitoring. This option provides more data to work with and may be more challenging, but strong analysis can lead to clearer and more well-supported conclusions.
>
> After choosing your land-based problem, on a separate document or on paper, answer each of the following questions using the associated spreadsheet (Learning Task Datasets file). Be sure to indicate which question you are answering.
>
> Framing the Problem
>
> Briefly describe your chosen issue and explain why it is relevant to Ontario's land and ecosystems.

After reviewing the data, write at least two clear, measurable questions you can ask to help guide your analysis?

Understanding the Data

Identify at least three important statistical calculations you will focus on and explain why each is useful for understanding the problem. (mean, median, range, variance, standard deviation)

Choose and justify at least two ways to organize or display the data (e.g., table, histogram, box-and-whisker plot, scatterplot). Explain why each representation is appropriate for your variables.

Statistical Analysis

Calculate the variables you identified in question 3. Show one worked example for each type of calculation you use.

Show your calculated data in a clear table. Label all columns and include units where appropriate.

Create at least two different graphs that support your argument (scatter plot, bar graph, histogram, boxplot, or pie chart). Explain why each graph is appropriate.

Interpreting the Evidence

Identify any trends, clusters, or outliers. Explain what they suggest about your topic.

What claims can you make about past, present, and future conditions based on your data? Clearly state any assumptions you make.

Discussing the Data

Discuss at least two strengths and two limitations of the dataset, and how they might affect your conclusions.

Reflection and Communication

Using what you have learned in this task, explain how the species in your chosen problem (invasive or at risk) could affect the survival of another species in the same ecosystem. Describe at least one way this relationship could cause a species to become at risk.

Reflect on what the data revealed that a simple description of the issue would not.

Write a brief summary of your findings so that anyone, regardless of math skill level, could understand your conclusions and evidence. Your summary should be between 200-400 words typed, or 3 pages handwritten.

Identify one area of statistics where your understanding could be improved.

Evaluation Task Dataset Prompt

Below is what I have for the Evaluation task. I want you to use this and make up a google spreadsheet that can be used by the student

In the learning task, you identified and justified why your chosen land-based problem is an issue. In this evaluation task, you will build on that work by justifying an appropriate restoration strategy. Below, you will find a table outlining two possible restoration strategies for each land-based problem (Blanding's Turtle and Emerald Ash Borer).

Your task is to review the two strategies for your selected problem and determine which one should be implemented. You are provided with data for each strategy; this data is representative of how similar strategies have been used at other affected sites in the past. You must use this evidence, along with the guiding questions below, to justify your recommended solution. Finally, you will present your argument using your choice of approved format (e.g., poster, video, presentation, conferencing etc.) so that a non-expert could understand.

Restoration Strategies

Blanding's Turtle

Habitat Expansion

Habitat expansion involves increasing the total area of land and wetlands protected for Blanding's turtles. By providing more suitable habitat, this strategy aims to support larger and more stable populations. Its effectiveness is evaluated by analyzing changes in turtle population size in relation to the amount of protected habitat and using the data to predict future population trends.

Habitat Connectivity

Habitat connectivity focuses on linking previously fragmented habitats through wildlife corridors. This allows turtles to move safely between areas for nesting, feeding, and overwintering. Success is measured by comparing population data from isolated habitats to connected ones and analyzing long-term trends, variability, and population stability.

Emerald Ash Borer

Containment Zones

Containment zones aim to slow or prevent the spread of the emerald ash borer by treating infected or high-risk areas with insecticides. Resources are concentrated where the risk of spread is greatest, and effectiveness is assessed by comparing infestation levels in treated zones versus untreated zones over time.

Removal and Replacement

This strategy focuses on removing ash trees that are already infected and replacing them with a variety of native tree species. The goal is to reduce the number of host trees available to the emerald ash borer while increasing forest diversity, which can make future infestations less severe. Success is typically evaluated by comparing infestation rates and forest composition before and after removal and replanting efforts.

On a separate document or on paper, answer each of the following questions using the associated spreadsheet (Evaluation Task Datasets file). Be sure to indicate which question you are answering.

Framing the Problem

How could the effectiveness of each strategy be measured?

How could you compare the effectiveness of each strategy?

Understanding the Data

Identify the variables you will use to compare the two strategies. Explain why these variables are relevant.

Organize the data in a clear way that allows for direct comparison between strategies (e.g., tables or grouped summaries).

Statistical Analysis

Calculate appropriate statistical measures (e.g., mean, median, range, IQR, standard deviation) for each strategy.

Show one fully worked example for each type of calculation used.

Create at least two graphs that support a comparison of the strategies. Explain why each graph is appropriate.

Interpreting the Evidence

Using statistical evidence, compare the short-term outcomes of the two strategies.

Use trends or patterns in the data to compare potential long-term outcomes of each strategy.

Identify any important trade-offs or real-world constraints (e.g., cost, variability, time, practicality) shown in the data.

Discussing the Data

Analyze the variability in the data for each strategy. How does this affect confidence in the results?

Discuss at least two strengths and two limitations of the data used in your comparison.

Based on the data, determine which strategy appears to be more effective or sustainable. Support your decision with statistical evidence.

Suggest at least one follow-up investigation or additional data that could improve future decision-making.

Reflection and Communication

Present your findings so that a non-expert can understand your conclusions and evidence? Justify your choice of format (e.g., report, poster, video, presentation, etc.).

After completing your presentation of findings, reflect on this task and describe one thing that went well, one thing that challenged you, and one thing you would have done differently if given the option to try again.

References

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